Seiberg-Witten $\Rightarrow$ Gromov

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Introduction

This report was put together for Math 242 at UC Berkeley during the spring of 2014. It contains a rough outline of the techniques to produce the Seiberg-Witten equations, Gromov-Taubes invariant, and the method used by Taubes to show that the Seiberg-Witten equations can produce the Gromov invariant.

A general outline of the history related to this exposition: In 1983, Donaldson [1] used techniques of gauge theory to explore the diffeomorphism type of 4-manifolds. By looking for instantons on four manifolds, Donaldson proved that the intersection form on a 4-dimensional manifold is diagonalizable. Corollaries of this theorem showed that there existed different differential structures for the same topological manifolds.

While the Donaldson invariants were very useful, they were difficult to compute. In particular, the moduli space of connections with self dual curvatures was not compact. In 1994, Seiberg and Witten used techniques they developed in [12] to develop the Seiberg-Witten invariants attached to Spin^c on 4-manifolds. The solutions to the Seiberg-Witten equations formed a smooth compact moduli space which captured the same data as the Donaldson invariants, but were easier to compute. The Seiberg-Witten invariant is an assignment to every Spin^c structure on a manifold a number that "counts" the number of solutions to the Seiberg-Witten equations.

Separately from Donaldson's work on instantons, Gromov in 1985 developed the theory of pseudoholomorphic curves for symplectic manifolds. In his paper [3], Gromov gave the basic ideas for "Gromov invariants". Given a symplectic manifold M with compatible J-holomorphic structure, Gromov studied the moduli space of pseudoholomorphic curves that represented a class in the 2nd homology of M. By requiring the pseudoholomorphic curves to pass through a certain number of points, one could make the moduli space a finite number of points and count pseudoholomorphic curves.

This brings us to the result that we are going to explore in this exposition. In a series of papers ([14], [13], [15], [16]) Taubes showed an equivalence between a Gromov-type invariant and the Seiberg-Witten invariants. Taubes used these results to compute the Gromov invariants for manifolds where the Seiberg-Witten invariants were understood. Interestingly, his paper shows that the Gromov-Taubes invariant, though apparently an invariant of symplectomorphism type, actually is only dependent on the diffeomorphism type of the manifold.

Here, we will look how solutions to the Seiberg-Witten equations can be used to produce pseudoholomorphic curves. The first two sections are the general background knowledge needed to state the problem, and short outlines of the proofs used in the construction of the Seiberg-Witten and Gromov-Taubes invariant. The sections most related to the course are Section 1.5 on page 10, and Parts 2 and 3. Of particular interest is Section 3.3 on page 30 which outlines the broader impacts of Taubes's work to symplectic geometry and J-holomorphic curves. A general outline of the main ideas of the paper are as follows

1. We start with an exploration of the Seiberg-Witten invariants.
    - Given a Spin Structure $\tilde{P} \to M$ (Section 1.1 on page 1) over a 4 dimension manifold with metric g, one can use the Dirac operator (Subsection 1.1.2) to set up an equation searching for positive spinors that satisfy a certain set of equations, the Seiberg-Witten Equations
    - By taking a slight perturbation η to the Seiberg-Witten Equations, the moduli space of the solutions $\mathcal{M}_{\mathrm{SW}}^*{}_\eta(\tilde{P})$ is a compact smooth manifold whose dimension can be computed with the Atiyah-Singer index theorem (Section 1.2 on page 3).
    - It can be shown that a homotopy of metric g induces a cobordism between associated moduli spaces. Therefore the cobordism type of $\mathcal{M}_{\mathrm{SW}}^*{}_\eta$ is independent of metric. By counting solutions by taking an appropriate integral over $\mathcal{M}_{\mathrm{SW}}^*{}_\eta(\tilde{P})$ of certain homology class, we get a number associated to every spin bundle that is independent of g. We can think of the Seiberg-Witten invariants therefore as an assignment
    $$SW : \mathrm{Spin}(M) \to \mathbb{Z}.$$

- We spend the last section exploring the Seiberg-Witten equations over symplectic 4 manifolds. Here, we show that there is a particularly nice choice of perturbation parameter that relies on the symplectic structure. Furthermore, by using the almost complex structure we are able to identify $\mathrm{Spin}(M) \simeq H_2(M, \mathbb{Z})$, and therefore can think of the Seiberg-Witten invariant as an assignment

$$SW : H_2(M, \mathbb{Z}) \to \mathbb{Z}.$$

2. The second chapter is used to look at pseudoholomorphic curves and Gromov-Taubes invariant.
   - We first explore the basics of psuedoholomorphic curves and their associated moduli spaces.
   - Given a class $A \in H_2(M, \mathbb{Z})$, we then look at $\mathcal{H}(A)$, the space of pseudoholomorphic submanifolds that pass through some specified number of generic points, modulo the case of multiply covered tori, which we will explore later.
   - We will not talk about the awfulness which is multiply covered tori.
   - The final section on Gromov-Taubes invariant is to count the number of curves that show up in $\mathcal{H}(A)$. We can therefore view Gromov-Taubes invariant as an assignment

$$Gr : H_2(M, \mathbb{Z}) \to \mathbb{Z}.$$

3. Finally, we would like to show some relations between these invariants. In the third part, we explore a small section of Taubes's proof: we construct a surface from the Seiberg-Witten equations. We leave off the proof that the surface that we have constructed is a submanifold or pseudoholomorphic, but show that the "interesting" behavior of the Seiberg-Witten solutions fall on a Hausdorff dimension 2 set. Our outline of this proof is roughly as follows:
   - By looking at a set of solutions to the perturbed symplectic Seiberg-Witten equations, we can create a sequence of currents on the manifold related to solutions of the Seiberg-Witten equations.
   - In Subsection 3.2.1, we study the energy of these currents, and show that they are bounded. In particular, we can find a convergent subsequence of these currents.
   - To better understand the behavior of these currents, we look at the zero set of the solution of the Seiberg-Witten equations.
   - In fact, most of the interesting behavior of a solution of the Seiberg-Witten equations is determined by this zero set (Subsection 3.2.2). We show that most characteristics of the Seiberg-Witten equations all drop off exponentially from the zero set of the solution when we vary the perturbation parameter.
   - We characterize the zero set of the Seiberg-Witten equations by a set of Hausdorff dimension 2 in Subsection 3.2.3. This completes the relation between the Seiberg-Witten equations and Gromov-Taubes invariant.

   Taubes later goes on to show that this set C is a psuedoholomorphic curve, and that it contains the relevant information to recover the Gromov Taubes invariant.

Resources

Our primary resources for this exposition are

- For the Seiberg-Witten invariants, [8], [4] and [9].
- For understanding psuedoholomorphic curves, besides notes from class, we used [6]. For the definition of Gromov-Taubes invariants, both [7] and [14] were used.
- The construction of psuedoholomorphic curves from Seiberg-Witten monopoles is from [15] and [13].

Notation

cl The Clifford multiplication map

$\mathcal{C}(\tilde{P})$ The SW configuration space

$\tilde{\mathcal{C}}(\tilde{P})$ The Reduced SW configuration space

$\mathrm{ev}(\phi, \tau, z_1 \dots z_l)$ The evaluation mapping

F_A^+ The self dual part of the curvature F_A

F_{SW} The Seiberg-Witten Function

$\mathcal{G}(\tilde{P})$ Gauge actions on SW configuration space

$Gr(A)$ The Gromov Invariant of A

J Almost complex structure

$\mathcal{L}$ The line bundle associated to a Spin structure

$\mathcal{M}_g(A)$ Moduli space of genus g pseudoholomorphic curves representing A

$\mathcal{M}_{\mathrm{SW}}(\tilde{P})$ The Seiberg-Witten moduli space

$\mathcal{M}_{\mathrm{SW}_\eta}(\tilde{P})$ η-perturbed redusce SW moduli space

$q(\phi)$ The quadratic map of a spinor

$\mathrm{Spin}(n)$ The n-dimensional spin group

$S_\pm$ The positive and negative spin representations

ω Symplectic Form

$W^{k,p}(E)$ Sobelov (k,p) sections of a bundle E

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The Seiberg-Witten Invariants

For our introduction to Seiberg-Witten invariants, we will first talk about the construction of Spin structures over a manifold, before moving onto the construction of the Seiberg-Witten equations. We then discuss the topology of the solution space of those equations. Our conclusion will explore some of the additional structure that a symplectic manifold can impose on the moduli space of solutions.

1.1 Spin Structures

Our exposition follows several parts:

- In Subsection 1.1.1 we will look at the spin group and its representations. In particular, we will look at splittings of the Clifford algebra and the action of the spin group on this.
- Having developed the representation theory of spin groups and Clifford algebras, Subsection **??** looks at principal spin bundles. We are interested in knowing the number of different principal Spin^c bundles there are over a manifold, and the natural connections that rise out of the metric structure on that bundle.
- Of particular interest to us is the fact that over the Spin bundles, the Laplacian has a square root, the Dirac Operator. Subsection 1.1.2 develops the basic properties of the Dirac operator, as well as giving it an explicit description.
- Finally, as the Seiberg Witten equations are define for four manifolds, we explore some nice properties of Spin bundles and Dirac operators in dimension 4.

1.1.1 Spin Groups and Spin Bundles

This is a quick overview of the terminology and notation that we will be using here; for an introduction to spin geometry we refer the reader to [18], [2], and [8].
Recall that given a metric structure on M, we can form the principal bundle of orthonormal frames which has an action on the tangent bundle via the standard representation. The group $SO(n)$ which acts on these frames has a double cover $\mathrm{Spin}(n)$ which acts on $SO(n)$ by conjugation. We would like to know when this map gives us principal $\mathrm{Spin}(n)$-bundle over a manifold M.

Definition 1.1.1

A **spin structure** on an oriented Riemannian manifold M is a principal $\mathrm{Spin}(n)$-bundle $\tilde{P} \to M$ along with a map to the frame bundle $\tilde{P} \to \mathrm{Fr}$ such that the following diagram commutes

$$\begin{array}{ccc} \tilde{P} \times \mathrm{Spin}(n) & \longrightarrow & \tilde{P} \\ \downarrow & & \downarrow \\ \mathrm{Fr} \times \mathrm{SO}(n) & \longrightarrow & \mathrm{Fr} \\ & & \downarrow \\ & & M \end{array}$$

In other words, a spin structure is a lift of the frame bundle to a representation of $\mathrm{Spin}(n)$ which is compatible with the double cover $\mathrm{Spin}(n) \to SO(n)$. If a manifold admits a spin structure, it is called **spinnable**.
As suggested by the definition, not every manifold admits a spin structure. The obstruction to the existence of a spin structure is given by this proposition.

Proposition 1.1.2

There are natural correspondence between the following three sets.

(a) A spin structure $\tilde{P} \to M$

(b) An element $\alpha \in H_1(\mathrm{Fr}(M); \mathbb{Z}_2)$ that restricts to the generator of each fiber.

(c) A trivialization of $\mathrm{Fr}(M)$ over the 2-skeleton

Not every four manifold admits a spin structure, but by adding an additional degree of freedom to the spin group we can form group which always admits this kind of desired lifting.

Definition 1.1.3

Define the group $\mathrm{Spin}^c(n)$ by

$$\mathrm{Spin}^c(n) = (\mathrm{Spin}(n) \times U(1))/\mathbb{Z}_2$$

Where $\mathbb{Z}_2$ is the subgroup $\{(1,1),(-1,-1)\}$ of $\mathrm{Spin}(n) \times U(1)$

Likewise, we can define a complex spin bundle over a manifold. A manifold is said to posses a Spin^c structure if there exists a Spin^c bundle whose representation over the frame bundle gives the same bundle structure as the orthogonal frame bundle. The extra degree of freedom in this bundle ensures in the four dimensional case there is always a lifting of a principal $SO(n)$ bundle to a principal Spin^c bundle.

Proposition 1.1.4

There exists a Spin^c structure whenever $c_1^{top}(\mathcal{L}) \cong w_2(M) \mod 2$.

Finally, the Spin^c structures on a manifold are a $H_2(M, \mathbb{Z})$ torsor.

Claim 1.1.5 There is a natural association between the Spin^c structures above M and $H^2(M, \mathbb{Z})$.

This identification can be seen by taking the associated line bundle $\mathcal{L} = \sqrt{\det S}$ which is associated to the bundle of Spinor fields. This associated line bundle is has a $U(1)$ gauge action which extends to a $U(1)$ gauge action on the complex spinor bundle.

From here on out, M will be a four dimensional real manifold. In the four dimensional case, we have an isomorphism between $\mathrm{Spin}(4) \simeq U(2) \times U(2)$, which gives us two complex representations of the spin group. We will call these the **positive and negative spinor representations** and denote then

$$\rho_\pm : \mathrm{Spin}(4) \to S_\pm \simeq \mathbb{C}^2$$

The groups $\mathrm{Spin}(n)$ and $\mathrm{Spin}^c(n)$ are subgroups of the Clifford algebra and its complexification. For us, it will be easiest to think of the Clifford algebra associated to the tangent bundle TM with inner product g by $\Lambda^* TM$ equipped with the Clifford multiplication structure defined on vectors by:

$$\mathrm{cl}(a)b = a \wedge b - g(a, b).$$

and extended by linearity. The Clifford multiplication structure induces a map between the positive and negative (complex) spin representations

$$\mathrm{cl} : S_+ \otimes TM \to S_-$$

$$\mathrm{cl} : S_- \otimes TM \to S_+$$

and satisfy the relation $\mathrm{cl}(v)^2 = -1$.

1.1.2 The Dirac operator

If M is a Riemannian manifold, it comes with the Levi-Civita connection ∇ attached to it. This naturally lifts to a connection on the real spinor bundles by pulling back the connection 1-form along its representation.

However, when we are working with complex spinors, we must also specify how to take derivatives along the added $U(1)$ direction in order to get a connection on complex spin bundles. Once we have picked a $U(1)$ connection A, we can define a connection ∇_A on the complex spinor bundle.

Proposition 1.1.6 Properties of the Spin Connection

The connection ∇_A has the following properties

1. It is unitary, that is if s_1, s_2 are two sections of $\tilde{P}$, we have that
$$d\langle s_1, s_2\rangle = \langle \nabla_A s_1, s_2\rangle + \langle s_1, \nabla_A s_2\rangle$$
2. It is a "module derivation": if α is a section of TM, and ϕ a section of S, then
$$\nabla_A(\alpha\cdot\phi) = \tilde{\nabla}_A(\alpha)\cdot\phi + \alpha\cdot(\nabla_A\phi)$$
where the $\tilde{\nabla}$ is the Levi-Civita connection, and the $\cdot$ is Clifford multiplication.
3. The $U(1)$ gauge action of σ on $\mathcal{L}$ is compatible with the action on the connection; that is
$$\tilde{\nabla}_{A\cdot\sigma} = \sigma^{-1}\tilde{\nabla}_A\sigma$$

Dirac operators are a kind of square root to a Laplacian. When a manifold comes with a Spin^c structure, there is a natural Dirac operator to consider.

Definition 1.1.7

Let $P \to M$ be a principal $SO(n)$ bundle, and let $S_{\mathbb{C}}(\tilde{P})$ be the associated (complex) spin bundle. Let $\tilde{\nabla}_A$ be the connection on $S_{\mathbb{C}}$ induced from the Levi-Civita connection and a fixed $U(1)$ connection A. Pick $\{e_i\}$ an orthonormal frame for TM_x, then define the **Dirac Operator** on spinors $\sigma(x)$ by

$$\begin{aligned}\not{\partial}_A : \Omega^0(M, S_{\mathbb{C}}(\tilde{P})) &\to \Omega^0(M, S_{\mathbb{C}}(\tilde{P}))\\ \sigma &\mapsto \mathrm{cl}(\sigma)\nabla_A(\sigma)\end{aligned}$$

where Clifford multiplication acts on $\nabla_A(\sigma)$ by using the metric.

This definition is independent of the choice of frame for TM_x. In local coordinates we can write Dirac multiplication as

$$\sum_{i=1}^{n} \mathrm{cl}(e_i)(\tilde{\nabla}_{e_i}(\sigma)(x).$$

We have the following facts about the Dirac operator

Proposition 1.1.8

If $\not{\partial}$ is a Dirac operator associated to a spin structure, then it is self adjoint $\not{\partial} = \not{\partial}^*$ and has the following Weitzenboöck presenation:

$$\not{\partial}^2 = (\nabla^E)^*\nabla^E + \mathrm{cl}(F(\nabla^E)) = \nabla_A^*\nabla_A + \frac{\kappa}{4} + \frac{F_1}{2}.$$

1.2 The Seiberg-Witten Equations

We now have the appropriate set up for the Seiberg-Witten equations. Our plan of attack is roughly:

- State the Seiberg-Witten equations, and provide some intuition for them. We'll also be interested in the differential of these equations, and presenting them as a functional.
- Look at the analytical considerations that need to be taken into account in order to topologize the space of Seiberg-Witten configurations.

- Show how the compatibility of the gauge group with the Seiberg-Witten equations allow us to remove the choice of $U(1)$ connection.

Definition 1.2.1 Seiberg-Witten Equations

Given $\tilde{P} \to M$ a Spin^c structure, the **Seiberg-Witten equations** are

$$\begin{aligned} F_A^+ &= q(\phi) \\ \not{\partial}_A \phi &= 0 \end{aligned}$$

where $\phi \in \Omega^0(M, S^+(\tilde{P}))$ is a positive spinor field for the orthogonal frame bundle $P \to M$ (and q and F_A^+ as defined below)

Let's explain what these equations mean.
According the physicists, the second equation means that we are looking for a field which represents a **Fermion** in quantum field theory. Mathematically, it is not too hard of an equation to understand— we are interested in an element of the kernel of the dirac operator.

The first equation is a little more complicated. The left hand side $F_A^+ = \frac{1}{2}(F_A + *F_A)$ represents the **self-dual** portion of the curvature two form. The right hand side is the **quadratic map**

$$\begin{aligned} q : \Omega^0(M, S^+(\tilde{P})) &\to \Omega^2(M, T_{\mathbb{C}}M) \\ \phi &\mapsto \phi \otimes \phi^* - \frac{|\phi|^2}{2}\,\mathrm{id} \end{aligned}$$

sending a section of the positive spinor bundle to a self dual two form. Notice that $\phi \otimes \phi^*$ is an endomorphism of the positive spinor bundle. There exists an isomorphism

$$\mathrm{End}_{\mathbb{C}}(S^+(\tilde{P})) \cong (Cl_0(P) \otimes \mathbb{C})^+ \cong \mathbb{C}\left(\frac{1+\omega_{\mathbb{C}}}{2}\right) \oplus (\Lambda^2_+(TM) \otimes \mathbb{C})$$

which tells us that $q(\phi)$ is a purely imaginary self-dual two-form. Here, $\omega_{\mathbb{C}}$ the Clifford product of all the elements of an orthonormal basis.

So given a fixed unitary connection A, a solution to the Seiberg-Witten equations corresponds to a spinor field whose quadratic part matches the self-dual portion of the curvature.
When we are finding solutions to the Seiberg-Witten Equations, we will be changing both the connection A and the spinor field ϕ. So in general we will want to look at pairs (A, ϕ) where A is a $U(1)$ connection on the determinant line bundle and ϕ is a positive spinor field. While we will eventually want smooth solutions to these equations, we will start with $W^{2,2}$ sections of these bundles to topologize the space of solutions; in the end, elliptic regularity will give us that all of our solutions are smooth.
Define $\mathcal{A}_{W^{2,2}}(\mathcal{L})$ to be the space of unitary $W^{2,2}$ connections on $\mathcal{L}$.

Definition 1.2.2 SW-Configurations

Let M be a riemannian manifold with spin structure $\tilde{P} \to M$ a spin bundle with determinant line bundle $\mathcal{L}$. Then the **configuration space** is defined to be

$$\mathcal{C}(\tilde{P}) = \mathcal{A}_{W^{2,2}}(\mathcal{L}) \times W^{2,2}(S^+(\tilde{P}))$$

Claim 1.2.3 The tangent space of the configuration space at a point can be identified with

$$W^{2,2}(T^*M \otimes i\mathbb{R}) \oplus S^+(\tilde{P})$$

In order to study solutions to the Seiberg-Witten equations, we define the **Seiberg-Witten function**

$$\begin{aligned} F_{\mathrm{SW}} : \mathcal{C}(\tilde{P}) &\to W^{1,2}\left((\Lambda^2_+ T^*M \times i\mathbb{R}) \oplus S^-(\tilde{P})\right) \\ (A, \phi) &\mapsto (F_A^+ - q(\phi), \not{\partial}_A(\phi)) \end{aligned}$$

and the **Seiberg-Witten Functional**

$$SW(A,\phi) = \int_M \left(|\not\partial_A \phi|^2 + |F_A^+ - q(\phi)|^2\right) dv$$

Notice that the solutions to the Seiberg-Witten equations that we have defined above are simply the configurations which have $F_{\text{SW}}(A,\phi) = 0$.

Lemma 1.2.4

The Seiberg-Witten function F_{SW} is a smooth mapping and its differential is described by the map

$$dF_{\text{SW}\,A,\phi} = \begin{pmatrix} P_+ d & -dq_\phi \\ \cdot\frac{1}{2}\phi & \not\partial_A \end{pmatrix}$$

where P_+ stands for the self-dual component.

The gauge group on the spin bundle gives us an action on the configuration space. In order for this action to be well defined, we need a little more control on the derivatives of the gauge transformations. We define the **space of gauge transformations** to be maps

$$\mathcal{G}(\tilde{P}) := W^{3,2}(M; S^1).$$

Given $\sigma \in \mathcal{G}(\tilde{P})$, define the right action of σ on $\mathcal{C}(\tilde{P})$ by

$$(A,\phi)\cdot\sigma = ((\det\sigma)^* A, S^+(\sigma^{-1}(\phi))).$$

Notice that the space of gauge transformation is a group under Sobelov multiplication. Furthermore, the $\mathcal{G}(\tilde{)})$ has the necessary regularity so that its action on $\mathcal{C}(\tilde{P})$ is well defined. Notice here that $\det(\sigma) = \sigma^2$. Furthermore, this action is going to play extremely nicely with the Seiberg-Witten function. There is also a way to define an action of $\mathcal{G}(\tilde{P})$ on the codomain of the Seiberg-Witten function.

Lemma 1.2.5

Let $\sigma \in \mathcal{G}(\tilde{P})$ be a gauge transformation. Then we have that

$$F_{\text{SW}}((A,\phi)\cdot\sigma)) = (F_{\text{SW}}(A,\phi))\cdot\phi$$

This gives us the nice result that the solutions to the Seiberg-Witten equations are invariant under the gauge transformations. So we can ignore the choice of unitary connection A by passing the quotient space.
In general, this quotient is not a smooth manifold, as some elements may have a nontrivial stabilizer in $\mathcal{G}(\tilde{P})$. However, there is a nice way to remove such elements. A configuration (A,ϕ) is called **irreducible** if $\phi \neq 0$. Let $\mathcal{C}_{irr}(\tilde{P})$ be the open subset of irreducible configurations.

Lemma 1.2.6

Let $(A,\phi) \in C_{irr}(\tilde{P})$ be an irreducible configuration. Then the stabilizer of (A,ϕ) in the gauge group is trivial.

If none of our solutions are reducible, then we can remove the reducible configurations from our configuration space.

Proposition 1.2.7 Removing Reducible Solutions

Suppose we have a spin structure $\tilde{P} \to M$, and we know additionally that

- $b_2^+(M) > 0$.
- $c_1(\mathcal{L})$ is not a torsion homology class.

Then for a generic metric on M, there are no reducible solutions to the Seiberg-Witten Equations

Definition 1.2.8 Quotient Configuration Space

Define the **quotiented Seiberg-Witten configuration space** to be

$$\tilde{\mathcal{C}}(\tilde{P}) := \mathcal{C}(\tilde{P})/\mathcal{G}\tilde{P}).$$

The irreducibe Seiberg-Witten configurations are defined to be

$$\tilde{\mathcal{C}_{irr}}(\tilde{P}) := (\mathcal{C}(\tilde{P}) \setminus \{\text{Reducible Configurations}\})/\mathcal{G}\tilde{P}).$$

The **Seiberg-Witten moduli space** associated to a Spin^c structure $\tilde{P} \to M$ is defined to be

$$\mathcal{M}_{\mathrm{SW}}(\tilde{P}) := F^{-1}(0) \subset \tilde{\mathcal{C}}(\tilde{P}).$$

As mentioned above, the obstruction from the quotient being a smooth manifold is these reducible configurations. While there are a lot of technical details to check (in particular, showing that $\tilde{\mathcal{C}}(\tilde{P})$ is Hausdorff) one can show the following:

Proposition 1.2.9

The reduced configuration space $\tilde{\mathcal{C}_{irr}}(\tilde{P})$ is a Hilbert manifold, with tangent space at $[A, \phi]$ isomorphic to

$$W^{2,2}\left((T^*M \otimes i\mathbb{R}) \oplus S^+(\tilde{P})\right) / \operatorname{Im}(2d, - \cdot \phi)$$

In the case where $b_2^+(M) > 0$ and $c_1(\mathcal{L})$ is not a torsion cohomology class, the Seiberg-Witten moduli space is the preimage of 0 in $\tilde{\mathcal{C}_{irr}}(\tilde{P})$.

1.3 The Seiberg-Witten Moduli Space

In this section, we will want to prove two fact about $\mathcal{M}_{\mathrm{SW}}(\tilde{P})$. We will make the assumption that $b_2^+(M)$ is greater than 0, to remove reducible solutions.

1. Our first goal will be to get an idea of the dimension of this space.
2. We will also want to show that this space is compact.

We will then look at a slight variation of the Seiberg-Witten Moduli space, to finally show

3. For generic perturbations, $\mathcal{M}_{\mathrm{SW}}^*{}_\eta(\tilde{P})$ is a smooth manifold

We will focus on proving the compactness and smoothness result. For dimension, we can use the Atiyah-Singer index theorem to prove the following claim.

Claim 1.3.1 Let $(A, \phi) \in \mathcal{M}_{\mathrm{SW}}(\tilde{P})$ be an irreducible solution. Then the dimension $\mathcal{M}_{\mathrm{SW}}(\tilde{P})$ around (A, ϕ) is given by

$$d = (c_1(\mathcal{L})^2 - 2\chi(M) - 3\,\mathrm{sgn}(M))/4.$$

1.3.1 Compactness of $\mathcal{M}_{\mathrm{SW}}(\tilde{P})$

Definition 1.3.2 Palais-Smale Condition

Let Y be a Hilbert manifold with the action of a Lie group G and a smooth functional $f : Y \to \mathbb{R}$ which is G-invariant. Then f is **Palais-Smale** if for any sequence $\{x_k\} \subset Y$ such that

1. $f(x_k)$ is bounded and
2. $df(x_k) \to 0$ as $k \to \infty$

there exists a subsequence $\{x_k'\}$ and a sequence $\{g_k\} \subset G$ such that $g_k x_k' \to x$ with $df(x) = 0$ and $f(x) = \lim_k f(x_k)$.

Here is the theorem that we want to prove.

Theorem 1.3.3 Compactness of $\mathcal{M}_{\mathrm{SW}}(\tilde{P})$

Given a sequence (A_n, ϕ_n) of solutions to the Seiberg-Witten Equations, there exist gauge transformations σ_n such that

$$\sigma_n(A_n, \phi_n)$$

has a smoothly convergent subsequence to a Solution of the Seiberg-Witten equations.

The meat of this prove relies on proving bounds on the Seiberg-Witten equations.

A priori bounds on curvature

To show compactness of the Moduli space of the Seiberg-Witten equations, we use a bound on the curvature. From an earlier section, we know that we can write the square of the Dirac operator in its Weitzenbock presentation Proposition **??**. We start by bounding the norm of the Seiberg-Witten equations, first by bounding the norm of F_A^+. From here, we can try and get a bound on the norms of the solutions.

Lemma 1.3.4
Curvature Identity to SW Equations

If (A, ϕ) is a solution to the Seiberg-Witten equations, then we have that

$$\|\nabla_A(\phi)\|_{L^2}^2 + \frac{1}{4}\langle\kappa\phi, \phi\rangle_{L^2} + \frac{\|\phi\|_{L^4}^4}{4} = 0$$

The proof is just a standard application of the Weitzenbock presentation, then you reexpress the curvature in terms of the spinor field using the Seiberg-Witten equation. We can naturally extend these estimates to create a bound for the Seiberg-Witten solutions in terms of the maximal curvature of M.

Lemma 1.3.5
Curvature Bounds for SW Equations

Let (A, ϕ) be a solution to the Seiberg-Witten equations. Then for every point $x \in M$ we have that

$$|\phi(x)| \leq \kappa_M^-$$

$$|F_A^+(x)| \leq \kappa_M^-/2$$

Where κ_M^- is the maximal curvature over all of M

Notice that if the maximal negative curvature is 0, there are no solutions to the Seiberg-Witten Equations.

... And now a bound on the norm

In general we are not able to show that the space of solutions to the Seiberg-Witten equations is compact. While the curvature is bounded, changes of gauge allow you to "escape" this compactness and find a set of solutions that do not converge. However, when quotient out by the action of the gauge group, we have enough symmetry to get compactness of the Seiberg-Witten moduli space. The argument to get a bound on A is gauge fixing argument and we only sketch an outline here. Fix a smooth connection A_0. Given any other connection A on $\mathcal{L}$, there exists a gauge transformation $\sigma \in \mathcal{G}(\tilde{P})$ such that

$$d^*(A - \sigma d\sigma - A_0) = 0$$

The coclosedness of the gauge allows us to pick a preferential gauge to work in; we will then bound the curvature with respect to that gauge. There exists a change of gauge with $d^*(\sigma(A) - A_0) = 0$ and

$$\|\sigma(A) - A_0\|_{W^{l,2}}^2 \leq C\|F_A^+\|_{W^{l-1,2}}^2$$

Putting together these bounds gives us the following restriction on the behavior of the Seiberg-Witten functional

Corollary 1.3.6
Seiberg-Witten Coerciveness [5]

The Seiberg-Witten functional is coercive, that is, there exists constant $c > 0$ such that

$$SW(A, \phi) \geq c^{-1}(\|\sigma^{-1}\phi\|_{W^{1,2}} + \|\sigma(A)\|_{W^{1,2}} - c$$

...and now putting it all together.

We are now ready to put together a proof of the convergence result.

Theorem 1.3.1. By the above section, we may assume that the sequence is that of coclosed one forms. This gives us that

$$\begin{aligned} d^* F_{A_n} =& (d + d^*) F_{A_n} \\ d^* d F_{A_n} =& (dd^* + d^* d) F_{A_n} \end{aligned}$$

By coercivity of the Seiberg-Witten functional, we may additionally assume that (up to another gauge transformation) that the $W^{l,2}$ norms of ϕ_n and $A_n - A_0$ are bounded for some fixed A_0. Since $Wl, 2$ embeds into C^{l-2}, we have that all of the continuous norms are bounded, and we may take a convergent subsequence. ■

One can modify this proof slightly to get that the Seiberg-Witten functional satisfies the Palais-Smale properties. As a corollary we get

Corollary 1.3.7 Let $\tilde{P} \to M$ be a spin-structure on a manifold. Then $\mathcal{M}_{\text{SW}}(\tilde{P})$ is compact.

1.3.2 The Perturbed Moduli Space

It isn't always the case that $\mathcal{M}_{\text{SW}}(\tilde{P})$ is a smooth manifold, in fact, we can only generically hope for it to be some choice of perturbation. In order to make this a smooth space we use the standard pertubation trick and a tranversality argument on generic sections. In order to have all of our derivatives work in the future, we will give ourselves a bit of extra regularity in the perturbation. Instead of working with the Seiberg-Witten functional, we will work with the perturbed Seiberg-Witten functional.

Definition 1.3.8 Define the (η)-**perturbed Seiberg-Witten Equations** to be the map

$$\begin{aligned} F : \mathcal{C}_4(\tilde{P}) \times W^{3,2}(\Lambda^2_+ T^* M \otimes \mathbb{R}) \to & W^{3,2}((\Lambda^2_+ T^* M \otimes \mathbb{R}) \oplus S^-(\tilde{P})) \\ F_\eta(A, \phi) \mapsto & (F^+_A - q(\phi) + i\eta, \not{\partial}_A(\phi)) \end{aligned}$$

where $\mathcal{C}_4(\tilde{P})$ is the configuration space with $W^{4,2}$ regularity. Let η be a self-dual real 2-form. The associated **Perturbed Seiberg-Witten Functional** is given by

$$SW_\eta(A, \phi) := \int_M (|\not{\partial}_A \phi|^2 + \frac{s}{4}|\phi|^2 + \frac{1}{2}|\frac{1}{2} q(\phi) - cl(i\eta)|^2 + |F_A + 2i\eta|^2$$

Redefine the **gauge group** to have extra $W^{5,2}$ Sobelov regularity to act on the Perturbed Seiberg-Witten configuration space.

By adding in this small perturbation parameter, we can use a Sard's theorem type of argument in the section parameter to find a zero section that is smooth.

Lemma 1.3.9 Suppose that (A, ϕ) is a solution to the η-perturbed Seiberg-Witten equation. Then at (A, ϕ, η) the differential DF is onto.

The surjectivity of this differential is the kind of result that will allow us to make a transveraslity argument for the intersection with the zero section to be a smooth manifold. Note in the case where we have $b^+_2(M)$ greater than 0, we had no reducible solutions.

Corollary 1.3.10 Let $\mathcal{M}_{\text{SW}-}(\tilde{P})$ be the space of irreducible perturbed solutions (up to gauge action) to the Seiberg-Witten equations. This space is a smooth submanifold of $\mathcal{C}_{irr,4}(\tilde{P})/\mathcal{G} \times W^{3,2}(\Lambda^2_+ T^* M \otimes \mathbb{R})$.

The proof of this corollary is by using the Atiyah-Singer index theorem to compute the index of the Fredholm operator associated to the elliptic complex given by the transveraslity result.

This space is far too large—it not only contains the solutions to the Seiberg-Witten equations, but a solution for every single perturbation. We would like to look at the solution above a single perturbation. This is going to be the final form of the moduli space of solutions that we are interested in .

Definition 1.3.11

We have a projection from $\mathcal{C}_{irr,4}(\tilde{P}) \times W^{3,2}(\Lambda^2_+ T^*M \otimes \mathbb{R}) \to W^{3,2}(\Lambda^2_+ T^*M \otimes \mathbb{R})$ which is a smooth mapping. We denote the fiber over a perturbation η by $\mathcal{M}_{\mathrm{SW}_\eta}$, the η**-perturbed moduli space**.

Theorem 1.3.12

For generic η, we have that $\mathcal{M}_{\mathrm{SW}_\eta}(\tilde{P})$ is a smooth submanifold of $C_{irr,4}(\tilde{P})/\mathcal{G}$ of dimension

$$\frac{c_1(\mathcal{L})^2 - 2\chi(M) - 6(\sigma(M))}{4}$$

Wrapping it up

Let's throw together all of the results from this section into one giant theorem.

Theorem 1.3.13
Properties of $\mathcal{M}_{\mathrm{SW}_\eta}$

Let M be a smooth oriented 4 Riemannian manifold with Spin^c structure $\tilde{P} \to M$. Additionally, suppose that $b_2^+(M) > 0$. Then for generic choice of η a self dual real two-form, we have that $\mathcal{M}_{\mathrm{SW}_\eta}(\tilde{P})$ is an oriented compact manifold of dimension

$$\frac{c_1(\mathcal{L})^2 - 2\chi(M) - 6(\sigma(M))}{4}.$$

With this in hand, we are ready to proceed to the production of the Seiberg-Witten invariants.

1.4 The Seiberg-Witten Invariants

With our construction of the perturbed moduli space to the Seiberg Witten equations, we are ready to actually go ahead and define the Seiberg-Witten invariants. Now that we have constructed these very nice moduli spaces $\mathcal{M}_{\mathrm{SW}}(\tilde{P})$, we would like to use them as invariants of a manifold M. However, these moduli spaces were constructed with a number of different choices, most importantly;

- The choice of Spin^c structure $\tilde{P}$
- The choice of metric g
- The choice of pertubation.

It turns out that only the first choice makes any difference, but the space of Spin^c structures is fairly nice. As mentioned before, the Spin^c structures form an $H_2(M, \mathbb{Z})$ torsor, so the Seiberg-Witten invariants that we are going to construct will give us an assignment

$$SW : \mathrm{Spin}^c(M) \to \mathbb{Z}.$$

However, the choices of g and of pertubation do change the structure of the moduli space. However, these changes will induce a cobordism on the moduli spaces of solutions, and so an invariant that captures the cobordism type of $\mathcal{M}_{\mathrm{SW}}(\tilde{P})$ will be our invariant.
How can we do this? We know that the Chern numbers of certain bundles over a manifold are dependent only on the cobordism type of the manifold; we therefore expect to look for things like this to define our Seiberg-Witten invariant. Our construction of the invariant will therefore come in two parts: showing that the moduli spaces of solutions actually are cobordant, and picking out an appropriate bundle to do this characteristic class construction.

1.4.1 The Invariant

There are several ways to pick different characteristic classes to integrate and construct an invariant, but this is the one that we are going to pick.

Definition 1.4.1

Let $\mathcal{M}_{\text{SW}}(\tilde{P})$ be the moduli space of solutions to the Seiberg-Witten eqautions, with $b_2^+ > 1$ so we have no reducible solutions.

- In the case where $\dim(\mathcal{M}_{\text{SW}}(\tilde{P})) < 0$, define $SW(\tilde{P}) = 0$.
- In the case where $\dim(\mathcal{M}_{\text{SW}}(\tilde{P})) = 0$, define $SW(\tilde{P})$ to be the signed count of points in $\mathcal{M}_{\text{SW}}(\tilde{P})$ with orientation.
- In the case where $\dim(\mathcal{M}_{\text{SW}}(\tilde{P})) > 0$, define

$$SW(\tilde{P}) := \int_{\mathcal{M}_{\text{sw}}} c_1(L)^{\frac{d}{2}}$$

where L is the determinant bundle given by the bundle $\mathcal{M}_{\text{SW}_-} \to \mathcal{M}_{\text{SW}_\eta}$.

These invariants are independent of choice of metric on M and pertubation η. Given two metrics g_0 and g_1 and pertubations, a generic homotopy between g_0 and g_1 induces a cobordism between the associated moduli space of solutions to the Seiberg-Witten equations. Since the integral of characteristic classes is dependent only on cobordism type, this shows that $SW : \text{Spin}^c \to \mathbb{Z}$ is a well defined invariant.

1.5 Seiberg-Witten over Symplectic Manifolds

In this section we explore how a J-holomorphic structure on a manifold induces adds additional structure to Spin structures and how we can express the Seiberg-Witten equations in these cases. A short over view of notation.

Definition 1.5.1

Let M^{2n} be a manifold. An **almost complex structure** on M is a bundle map $J : TM \to TM$ such that $J^2 = -1$. . We say that J is compatible with a metric g on M if J is an orthogonal transformation on the tangent spaces with respect to g.
This gives us a decomposition of the complexified exterior bundle into holomorphic and antiholomorphic forms.

$$\Omega^k M \otimes \mathbb{C} = \bigoplus_{i=0}^{k} \bigwedge^i T^{1,0} \otimes \bigwedge^{k-i} T^{0,1}$$

which is now equipped with a holomorphic and antiholomorphic differential, d and $\bar{d}$.

One interesting property of the complex valued forms is the amount of flexibility we have for raising and lowering the degree of forms. On the one hand, we can raise the degree of a form by multiplication. In the other direction, we can lower the degree of a form either by taking interior multiplication with the metric, or by taking on the of the canonical projections to the holomorphic or antiholomorphic parts, it turns out that it is possible to encode the Clifford multiplication structure by using these operations.

Proposition 1.5.2

If (M, g) is a Riemannian manifold equipped with compatible J-holomorphic structure, then we can create a spin structure with the following

- The Spin bundle is identified with the complex $(0, q)$ forms.
- The positive spinor representation is identified with the even $(0, 2q)$ forms.
- The negative spinors are identified with the odd $(0, 2q + 1)$ forms.

We additionally can explicitly express the Clifford multiplication of TM on $\Lambda_{\mathbb{C}}^{(0,q)}T^*M$ as follows:

$$v \cdot \mu = \sqrt{2}(\pi^{0,1}(v^*) \wedge \mu - \iota_{\pi^{0,1}(v^*)}\mu)$$

This happens to be the spinor representation associated with the canonical bundle. One can check in this case, the Dirac operator can be replaced with

$$\not{\partial}_A(\sigma)(p) = \sqrt{2}(\bar{\partial}(\sigma)(p) + \bar{\partial}^*\sigma(p)).$$

Furthermore, we can represent spinor fields σ as (α, β) where α is a section of $\Omega^0(M)$ and β is a section of $\Omega^{0,2}(M)$.
This gives us one example of a spin structure over a manifold with almost complex structure; how do we get the other ones? Given a Spin^c bundle, we can take describe the positive spinor bundle as follows:

Proposition 1.5.3

Let $P \to M$ be the Spin^c structure associated to complex line bundle E. Then we can identify the positive spinors with

$$S_+ = E \oplus (K^{-1} \otimes E)$$

If we pick a section u_0 of the canonical bundle, then we can pick a compatible spin covariant derivative A_0 by taking $D_{A_0}u_0 = 0$. We have from before that the set of Spin^c structures is a $H^2(M, \mathbb{Z})$ torsor. However when we pick out a canonical element of the Spin^c structures, we can actually identify the set of spin structures with $H^2(M, \mathbb{Z})$. We should therefore interpret the Seiberg Witten invariants not a function from Spin structures to $\mathbb{Z}$, but instead as

$$SW : H^2(M, \mathbb{Z}) \to \mathbb{Z}.$$

The symplectic form also allows for a nice choice of perturbation factor η. We can give a real parameterization to a perturbation factors by writing them as

$$\eta = -\frac{i \cdot r}{4}\omega + F^+_{A_0}$$

where A_0 is the canonical connection associated to the bundle. With this choice of perturbation, we have that the Seiberg-Witten equations are given by

$$\begin{aligned} \not{\partial}(\phi) &= 0 \\ F^+_A &= F^+_{A_0} + q(\phi) - ir\omega \end{aligned}$$

If we choose to represent the spinor field in terms of the decomposition of Proposition 1.5 then we can write

$$\phi = r^{\frac{1}{2}}(\alpha u_0 + \beta)$$

where α is a section of E and β is a section of $K^{-1} \otimes E$. Then we can write the Seiberg-Witten equations as follows

Claim 1.5.4 The solutions to the perturbed Seiberg Witten equations are equivalent to finding solutions for a parameter r to the symplectic Seiberg Witten equations.

$$\begin{aligned} \sigma(u_0 \otimes \nabla_a \alpha) + D_A\beta &= 0 \\ F^+_a = \frac{-i}{8}r(1 - |\alpha|^2 + |\beta|^2) \cdot \omega + \frac{ir}{4}(\alpha\beta^* + \alpha^*\beta) & \end{aligned}$$

We therefore can define the **r-perturbed moduli space of solutions** to be $\mathcal{M}_{\mathrm{SW}r}(\tilde{P})$. This is a smooth compact manifold for generic r.

Taubes's Gromov Invariant

In this exposition, we will take the approach used by Taubes in his paper [14].
Given a symplectic manifold M, one can look at the number of pseudoholomorphic curves which intersect a specified homology cycle in M. This general idea gives you a mapping from $\Phi : H_2(M, \mathbb{Z}) \to \mathbb{Z}$ by taking a signed count of these curves. There are some principle differences between Taubes's Gromov invariant and the Gromov-Witten invariant. The biggest change is the Taubes considers pseudoholomorphic submanifolds independent of parameterization. Here is an outline of how we define Taubes's Gromov Invariant

- We start with a bit of background on pseudoholomorphic curves. Section 2.1 covers the basic definitions and constructions. Of particular interest is the construction of the moduli space of J holomorphic curves, and the evaluation map of this moduli space. We show that this map is a pseudocycle and therefore represents an element of homology. This construction allows us to count the number of pseudoholomorphic curves that pass through k generic points.
- In Section 2.2 on page 20, we define Taubes's Gromov invariant, modulo some cases involving counting multiply covered torii. The majority of this section is devoted to showing that it is well defined.
- We leave off the more complicated case of counting multiply covered torii.

2.1 Background on Pseudoholomorphic Curves

McDuff has a number of great resources for learning more about Pseudoholomorphic curves, our primary reference here is [7] and [6]. Before we can get into developing Taubes's Gromov invariant, we start with some basic definitions for Pseudoholomorphic curves.

2.1.1 Almost Complex Structures

Definition 2.1.1

Let (M^{2n}, ω) be a symplectic manifold with almost complex structure J. Then we say that J is ω-**compatible** if

1. $g(v, w) = \omega(v, Jw)$ is a Riemannian metric.
2. $\omega(Jv, Jw) = \omega(v, w)$

If J satisfies the weaker property that $\omega(v, Jv) > 0$ whenever $v \neq 0$, then we say that J is ω-**tame**.

For example, $M = \mathbb{C}^n$ with the standard two form and standard J-structure has ω-compatibility.

Exercise 2.1.2 For any $p \in M$, we can choose a basis for T_pM in which ω, J, g are standard.

Notice that property 2 gives us that g is symmetric.

$$\begin{aligned} g(w, v) &= \omega(w, Jv) \\ &= \omega(-J^2 w, Jv) \\ &= \omega(-Jw, v) \\ &= \omega(v, Jw) \\ &= g(v, w) \end{aligned}$$

When we have that J is ω-tame, we can get a Riemannian metric by taking

$$g(v, w) = \frac{1}{2}(\omega(v, Jw) + \omega(w, Jv))$$

Notice in the case where J is ω compatible this simplifies to the metric that we had above. When we are talking about J-holomorphic curves, we will want to use (tame or) compatible J.

Proposition 2.1.3 The set of ω-compatible almost complex structures J on M is nonempty and contractible. Moreover, any ω-compatible almost complex structure or homotopy of ω-compatible almost complex structures on a compact subset of M can be extended over M.

This follows from the following lemma:

Lemma 2.1.4 Let (V,ω) be a symplectic vector space. Let $\mathcal{J}_c(V,\omega)$ be the linear maps J such that $J^2 = -1$ and J is ω-compatible. Then $\mathcal{J}_c(V,\omega)$ is contractible.

Proof. Done in the class! ∎

The proposition also holds with "compatible" replaced by the word "tame".

Exercise 2.1.5 If $\dim V = 4$, then g comes from an ω compatible almost complex structure J if and only if ω is self-dual with respect to g and $|\omega| = \sqrt{2}$.

2.1.2 J-holomorphic Curves

Definition 2.1.6 A **Riemann Surface** is a closed, connected two dimensional manifold Σ with an almost complex structure j. The **genus** of Σ is the number of holes in the surface.
Given (M,ω,J), a **J-holomorphic curve**[1] $(\Sigma,j) \to (M,J)$ is a smooth map $u:\Sigma \to M$ such that

$$J \circ du = du \circ j$$

People go bananas over pseudoholomorphic curves. Here are a few examples.

Let (M,ω,J) be symplectic manifold with almost complex structure.

1. The constant map is a pseudoholomorphic curve.
2. If $\Sigma \subset \mathbb{CP}^2$ is a nonsingular algebraic curve then the inclusion map is holomorphic.
3. If $M = \Sigma \times N$ and $J_m = j \oplus J_N$ for each $p \in N$, then $\Sigma \to M$ is holomorphic.

In general we are going to be interested in counting the number of pseudoholomorphic curves that pass through some prespecified number of sets.

Definition 2.1.7 Let (M,ω,J) be a compact symplectic manifold. Let $A \in H_2(M)$. Then define the **moduli space of genus g J -holomorphic curves** in A to be

$$\mathcal{M}_g(A) := \{u:\Sigma \to M | u \text{ is } J\text{-holomorphic}\}$$

where Σ is a closed Riemannian surface with almost complex structure j, and $u:\Sigma \to M$ a map with $J \circ du = du \circ j$ and $u_*[\Sigma] = A$.

It is tricky to count the curves in $\mathcal{M}_g(A)$, because you can sometimes take a curve and then get a new one by factoring through a branched cover. Clearly, these should be counted in a different way, otherwise you could build an infinite number of such curves by simply taking more and more branched covers.

What do we know about $\mathcal{M}_g(A)$ in general? We have this general properties of J holomorphic curves,

[1] Also called a pseudoholomorphic curve or holomorphic curve

Theorem 2.1.8 Adjunction Formula

If C is a genus g holomorphic curve in the class of A, then

$$A \cdot A = c_1(A) + 2(g-1) + 2m$$

where m is the number of double points.

Proof. First let us suppose that C is immersed. Then we have that the first Chern class of A is given by

$$\begin{aligned} c_1(A) =& c_1(TM, J)(C) \\ =& c_1(TC)(C) + c_1(NC)(C) \end{aligned}$$

The Chern class is related to the genus, while generic sections of the normal bundle has zeros related to the homology class that C represents, and the number of double points of the map C

$$= 2 - 2g + C \cdot C - 2m$$

■

While the general moduli space of parameterized J-holomorphic curves is a little difficult to get a hand on because many times the parameterization can be factored through another parameterization. We are not interested in such representations, so we will simplify by discounting them.

Definition 2.1.9

A holomorphic map $i : (\Sigma, j) \to (M, J)$ is **somewhere injective** if there exists $z \in \Sigma$ such that $u^{-1}(U(z)) = \{z\}$ and $du : T_z\Sigma \to T_{u(z)}M$ is injective.
We say that $u : (\Sigma, j) \to (M, j)$ is **Multiply covered** if there exists a branched over $\rho : \Sigma \to \Sigma'$ of degree $d > 1$ and a holomorphic map $u' : (\Sigma', j') \to (M, J)$ such that $u = u' \circ \rho$

In other words, a map is multiply covered if you can factor it through a branched cover, and somewhere injective means that there is a disk for which your map is an embedding. Notice that the image of every multiply covered curve can be represented by the somewhere injective curve that it factors through, therefore if we are only interested in the number of unparameterized J-holomorphic curves, we may "throw out" all of the multiply covered parameterizations.

2.1.3 The moduli space of somewhere injective holomorphic curves

Introduction

To prepare to understand the overall story, it would be better for you to know Fredholm theory and Atiyah-Singer index theorem (or Riemann-Roch theorem). Basically, what we are going to discuss is that

$$\mathcal{H}(A, J)$$

is a smooth manifold of dimension $n(2-2g)+2c_1(A)$ where A is an element of $H_2(M; \mathbb{Z})$.

Definition 2.1.10 The moduli space of somewhere injective J-holomorphic curves with 2nd homology A

$$\mathcal{H}(A, J) = \{u : \Sigma \to M \mid u(\Sigma) \text{ is somewhere injective } J\text{-holomorphic and } u_*[\Sigma] = A \in H_2(M; \mathbb{Z})\}$$

Fredholm theory

One of main ingredients for studying the moduli space $\mathcal{H}(A, J)$ is Fredholm theory. Let's see what Fredholm theory is, briefly. Suppose that we have an operator $f : X \to Y$ where X and Y are Banach spaces. A **Fredholm operator** is a *bounded linear operator* whose kernel and cokernel are *finite-dimensional* and *range is closed*. For such an operator we define **the index of the operator** $\operatorname{ind} f$ by

$$\operatorname{ind} f = \dim \ker f - \dim \operatorname{coker} f.$$

A fredholm operator $\bar{\partial}_J$

Let's first think about what $\mathcal{H}(A,J)$ is. Basically, it is a subset of all smooth maps $u:\Sigma\to M$ such that u is J-holomorphic. It means that du has no complex anti-linear part. Hence, if we define a map $\bar{\partial}_J$ which represents the complex anti-linear part of du then $\mathcal{H}(A,J)$ will be $\ker\bar{\partial}_J$. From here, we can get some help from Fredholm theory! So, let's move on to defining an appropriate Fredholm operator $\bar{\partial}_J$.

Let's define $\mathcal{M}(A)$ as a space of all smooth maps $u:\Sigma\to M$ which are somewhere injetive and $u^*[\Sigma]=A\in H_2(M;\mathbb{Z})$. This space can be interpreted as an infinite dimensional manifold of which tangent space at each point (which is a function u) is the space of all smooth vector fields on $u(\Sigma)$,

$$T_u\mathcal{M}=C^\infty(u^*TM).$$

Next, consider a vector bundle $:\mathcal{E}\to\mathcal{M}$ such that

$$^{-1}(u)=\{\text{smooth } J\text{-anti-linear 1-forms with values in } u^*TM\}.$$

Then, we have a section $\bar{\partial}_J:\mathcal{M}\to\mathcal{E}$ which maps u to the complex anti-linear part of du. As we have discussed just before, $\mathcal{H}(A,J)=\bar{\partial}_J^{-1}(0)$.

We want $\mathcal{H}(A,J)$ to be a manifold and it requires $\bar{\partial}_J$ be transversal to the zero section. For the next, let's define $D_u=\pi_u\circ d\bar{\partial}_J:T_u\mathcal{M}\to T_{(u,\bar{\partial}_J(u)=0)}\mathcal{E}\to{}^{-1}(u)$ so that D_u is an operator from $C^\infty(u^*TM)$ to the space of complex anti-linear 1-forms with values in u^*TM. Then the trasversality condition is equivalent to that D_u is surjective at the point u where $\bar{\partial}_J(u)=0$.

Basically, through computation, we can figure out that D_u is an elliptic first order partial differential operator and such kind of operator is always Fredholm.

From PDE theory, we have a series of claims about this operator

Claim 2.1.11 Consider the operator $D_u:C^\infty(u^*TM\to\Omega^{0,1}(u^*TM)$ as above.

- D_u is an elliptic first order PDO.
- $C^\infty(u^*TM)$ and $\Omega^{0,1}(u^*TM)$ are complex vector spaces.
- The antilinear part of D_u is order 0, so D_u has the same index as a Cauchy-Riemann operator.
- The set of almost complex structures J such that D_u is onto is generic.

Also we have a lemma stated as below.[6]

Lemma 2.1.12 Let $u:\Sigma\to M$ be a smooth map and $Q_u:C^\infty(\wedge^{0,1}T^*\Sigma\otimes_J u^*TM)\to C^\infty(u^*TM)$ be a right inverse of D_u such that

$$\|Q_u\|\le c_0,\quad \|du\|_{L^p}\le c_0,\quad \|\bar{\partial}_J(u)\|_{L^p}\le\delta$$

with respect to a metric on Σ such that $\mathrm{vol}(\Sigma)\le c_0$. Then for every $\xi\in\ker D_u$ with $\|\xi\|_{L^p}\le\delta$ there exists a section $\hat{\xi}=Q_u\eta\in C^\infty(u^*TM)$ such that

$$\bar{\partial}_J(\exp_u(\xi+\hat{\xi}))=0,\qquad \|\hat{\xi}\|_{C^\infty}\le c\|\bar{\partial}_J(\exp_u(\xi))\|_{L^p}.$$

(Here, p can be any value > 2.)

In words, what the lemma implies is that for a given small section ξ of $C^\infty(u^*TM)$ which is in the kernel of D_u, we can find a unique $\hat{\xi}\in\mathrm{im}D_u^*$ such that $\xi+\hat{\xi}$ ($\in$the tangent space $T_u\mathcal{M}$) makes u to be a J-holomorphic curve via the exponential map defined on $\mathcal{M}$. In mathematical expression, we have found a smooth map

$$\ker D_u\to\mathrm{im}D_u^*$$

which help $\xi\in\ker D_u$ make u a J-holomorphic curve.

The next lemma we would like to use is

Lemma 2.1.13

Under the same assumption as before with

$$||Q_u|| \leq c_0, \qquad ||du||_{L^p} \leq c_0,$$

if $v_0 = \exp_u(\xi_0)$ and $v_1 = \exp_u(\xi_1)$ are J-holomorphic curves such that $\xi_0, \xi_1 \in C^\infty(u^*TM)$ satisfy

$$||\xi||_{W^{1,p}} \leq \delta, \qquad ||\xi||_{W^{1,p}} \leq c_0,$$

and

$$||\xi_1 - \xi_0||_{L^p} \leq \delta, \qquad \xi_1 - \xi_0 \in \mathrm{im} Q_u,$$

then $v_0 = v_1$.

Using this lemma, we can now conclude that every J-holomorphic curve v, which is bounded in $W^{1,p}$ and sufficiently close to u can be uniquely expressed as

$$v = \exp_u(\xi + \hat{\xi})$$

for some $\xi \in \ker D_u$. So far we have just given a local chart from $\ker D_u$ to $\mathcal{H}(A, J)$ so that $\mathcal{H}(A, J)$ is a manifold. Now, we get the dimension of $\mathcal{H}(A, J)$ agreeing with $\dim \ker D_u$ which is finally $\mathrm{ind} D_u$ since D_u is onto. Now, by Atiyah-Singer index theorem or Riemman-Roch theorem, $\dim \mathcal{H}(A, J) = \mathrm{ind} D_u = n(2-2g) + 2c_1(u^*TM)$.

For smoothness of the manifold we can use this argument : for a generic choice of J, we have $\bar{\partial}_J^{-1}(0)$ a smooth manifold by the infinite dimensional version of Sard's Theorem (Sards-Smale Theorem). Therefore, the moduli space $\mathcal{H}(A, J)$ is a smooth manifold of dimension $n(2-2g) + 2c_1(u^*TM)$.

Definition 2.1.14

Define the moduli space of somewhere injective holomorphic curves to be

$$\mathcal{M}_g^*(A) = \{\phi \in Hom(\Sigma_g, M) \times \mathcal{T}_g \mid (*)\}$$

where the conditions $(*)$ are

- If we give Σ the complex structure τ, then $\phi : \Sigma_g \to M$ is J-holomorphic.
- The map ϕ is somewhere injective.
- $\phi(\Sigma)$ represents the homology class A.

This space is much nicer to work with. The benefit of using the multiply covered curves is that we (generically) get a manifold out of this moduli space. If we don't use these equivalence classes, these moduli spaces look like manifolds of various dimension that touch each other— with each component corresponding to curves of different multiplicity in branching. It should be noted that we introduce a different problem at tis point, in that some sequences of parameterizations of tori can converge to a multiply covered torii, but we will not discuss that at all.
We would like to study properties like the dimension of this manifold– before we can get there, we should look at the ways that we can represent different elements of $\mathcal{M}_g^*(A)$ by reparameterization.

Definition 2.1.15

Given a Riemann surface Σ_g, let $\mathcal{T}_g$ denote the **Teichmüller space** of Σ_g, the set of all complex structures taken up isotopy.

It is a common result [10] that the Teichmüller space of a manifold has dimension

$$\dim_{\mathbb{R}} \mathcal{T}_g = 6g - 6$$

and that we have a smooth mapping from the Teichmüller pace to the set of j-holomorphic structures on $\mathcal{J}$.

Definition 2.1.16

Given a Riemann surface Σ_g, let G_g be the reparameterization group. We can actually compute these explicitly to begin with:

- if $g = 0$, we have that reparameterization is given by $\mathrm{PSL}(2,\mathbb{C})$
- if $g = 1$, we have that we can reparameterize by $\mathrm{SL}(2,\mathbb{Z})$ extended by the torus.
- otherwise, G_g is the mapping class group $\pi_0(\mathrm{Diff}(\Sigma))$.

Theorem 2.1.17

For generic ω-tame J, the moduli space $\mathcal{M}_g^*(A)$ is an oriented manifold of real dimension

$$\dim_{\mathbb{R}}(\mathcal{M}_g^*(A)) = 2(c_1(A) + g - 1) + \dim G_g$$

Proof.

$$\dim \mathcal{M}_g^*(A) = \dim \mathcal{H}(A,J) + \dim \mathcal{T}_g + \dim G_g = 2c_1(A) + 2g - 2 + \dim G_g$$

■

Corollary 2.1.18

If $l = l_g(A) = c_1(A + g - 1)$, there is a evaluation mapping

$$\begin{aligned} \mathrm{ev} : \mathcal{M}_g^*(A) \times_{G_g} \Sigma^l &\to M^l \\ (\phi, \tau, z_1, \ldots z_l) &\mapsto (\phi(z_1), \ldots, \phi(z_l)) \end{aligned}$$

between manifolds of equal dimension $4l$.

Proof. This is just a dimension count. ■

What we would like now is a statement along the lines of "the number of curves (counted with sign) that pass through l generic points is given by the degree of the mapping ev." This would tell us that the choice of l points is not consequential and that we've actually defined an invariant of the manifold. However, the space $\mathcal{M}_g^*(A)$ is not compact, and therefore the degree of the mapping is not well defined. The failure to $\mathcal{M}_g^*(A)$ being compact is due to the possible convergence of J-holomorphic curves to "cusp curves" This is summed up in the following theorem

Theorem 2.1.19
Gromov's Compactness

Assume that (M,ω) is compact, and let J_i be a sequence of J holomorphic structures compatible with ω. Then any sequence of J_i holomorphic spheres converges weakly up to cusp curves.

In order to actually study this manifold $\mathcal{M}_g^*(A)$ properly, we will want to work with its compactification, which may include cusp curves. Then we will be able to introduce an idea of degree which is compatible with our notion of degree for maps but works for compact manifolds with boundary.

2.1.4 Cusp Curves

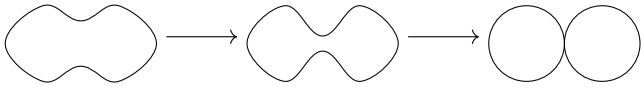
For this section we will restrict to looking at the moduli space of genus 2 curves.

Definition 2.1.20

Let (M,ω) be a compact symplectic manifold with ω-tame almost complex structure J. A **cusp curve** is a connected union $C = \bigcup_{i=1}^n C^j$ of J holomorphic spheres.

We can think of cusp-curves as being like a necklace of spheres that are just about to come apart. The reason to introduce cusp curves is to compactify the space of J-holomorphic curves. For instance, one can take the

curves converging as follows:



The curves on the left may be J-holomorphic, but the curve on the right is a cusp curve.
We can parameterize cusp curves in two different ways. The first is to provide a J-holomorphic parameterization for each sphere on the curve. In this way, we represent the cusp curve as a collection of functions $u^i : \mathbb{CP}^1 \to M$.
A second way to parameterize the cusp curve is with a single function. Notice that every singe cusp curve can still be parameterized as a smooth map from a connected domain to M, but this map will necessarily not be J-holomorphic.
Both of these characterizations of cusp curves is useful in defining what it should mean for a series of J -holomorphic curves to converge to a cusp curve.

Definition 2.1.21

Let $u_j : \mathbb{CP}^1 \to M$ be a sequence of J-holomorphic curve, $u : \mathbb{CP}^1 \to M$ be a smooth parameterization of the cusp curve, and $u^i : \mathbb{CP}^1 \to M$ be its associated J-holomorphic parameterization. Then we say that u_j converges to u if it converges in both of the following senses:

- (J-holomorphic convergence) We have associated fractional linear transformations ϕ_j^i and finite sets X^j such that $u_i \phi_i^j$ converges to ϕ_i uniformly on all derivatives over all compact sets of $\mathbb{CP}^1 \setminus X^j$.
- (Smooth Convergence) There exists diffeomorphisms $f_i : \mathbb{CP}^1 \to \mathbb{CP}^1$ such that $u_i f_i$ converges to u in the sup-norm topology.

We should expect that the space of cusp curves that pass through k generic points to have real dimension 2 less than the space of J-holomorphic curves that pass through k points, because we are imposing the additional condition on these curves that they are also connected to each other. The second reason that we would expect the cusp curves to have a smaller dimension than the pseudoholomorphic curves is because we think of them in some way as the limit of pseudoholomorphic curves that are about to bubble off. We will represent these cusp curves by the homology class that each bubble has along with which bubbles it is connected to; this is called the framing data and is denoted

$$D = (A^1, \dots A^N, j_2, \dots j_N)$$

where the A^i denotes the homology class of the ith bubble in the cusp curve, and j_i denotes that the i and j bubbles share a point in common.

Theorem 2.1.22

Let $\mathcal{M}(D, J)$ be the moduli space of cusp curves that fit the framing D. For generic J we have the dimension of this space is

$$\mathcal{M}(D, J) = 2 \sum_{j=1}^{N} (c_1(A^j) + 2n + 4(N-1)$$

While $\mathcal{M}(D, J)$ is not codimension 2 with respect to $\mathcal{M}(A, J)$, when we look at the evaluation map, we notice that the fiber product will cut down the dimension a lot. Therefore when we look at unparameterized curves, we have that:

Corollary 2.1.23

In the case of dimension four, the space of cusp curves is codimension 2 in the compactification of $\mathcal{M}_g(A) \times_G \Sigma^l$.

This means that the "boundary" of the compactification of moduli space of pseudoholomorphic curves is not very big; this stratification will be extremely useful for us when we want to talk about the homological properties of the evaluation map.

2.1.5 Evaluation Mapping and Pseudocycles

In order to understand the number the number of pseudoholomorphic curves that have image containing a specified number of generic points, we will look at the evaluation mapping for Pseudoholomorphic curves.

Theorem 2.1.24 The evaluation map represents a class in homology, whose multiplicity counts the number of curves passing through a generic point in M^l.

This will give us the corollary that allows us to count these curves.

Corollary 2.1.25 Given a generic set of points $(x_1, \dots x_l) \in M^l$, the preimage $\mathrm{ev}^{-1}(x_1, \dots, x_l)$ is finite. If A is sufficiently nice [2]then the number of points in the inverse image is independent of choice of J.

Since the manifold $\mathcal{M}(A, J, g)$ is only compact with the addition of cusp curves, we cannot use the usual degree of this mapping to count the number of points in the preimage of a generic point in M^l. However, we can get a very similar result by introducing the concept of a pseudocycle [7].

Definition 2.1.26 Pseudo-cycles

Let M be a compact manifold. A k-dimensional pseudo-cycle in M is a smooth map $f : V \to M$ where V is k dimensional and the boundary of $f(V)$ is of dimension at most $k - 2$.

Lemma 2.1.27 In dimension 4, The evaluation map is a Pseudocycle

Proof. This is immediate from 2.1.4. ∎

Definition 2.1.28 Let $f_0 : V_0 \to M$ and $f_1 : V_1 \to M$ be two k dimensional pseudocycles. Then we say that they are **bordant** if there exists a $(k+1)$ dimensional oriented manifold W with $\partial W = V_1 - V_0$ and an extension of f_0 and f_1 to $F : W \to M$.

Claim 2.1.29 If M is a compact smooth manifold of dimension greater than three, every element of $H_2(M, \mathbb{Z})$ can be represented by an embedded surface.

We will first show that pseudocycles naturally represent classes of cohomology, and then take the Poincaré dual.

Lemma 2.1.30 Let e and f be pseudocycles of complementary dimension. Then the intersection number of e and f depends only on the bordism class of e and f.

Proof. What we will show is that if f is bordant to $\varnothing$, then the intersection number of e and f is zero. Let F be a bordism between f and the empty set. This bordism can be chosen to be transverse to e, so the intersection of F and e is a 1 dimensional manifold. Then the intersection of e and f is given by the boundary of $F \cap e$. The number of signed points in the boundary of a 1 dimensional manifold is zero. ∎

We can use this property to define a class of cohomology for a pseudocycle e. Let e be a $(n-d)$ pseudocycle, and let $\alpha \in H_d(M, \mathbb{Z})$ be a class of homology. Then let a be a cycle which represents the element α, and define

$$\Phi_{[e]}(\alpha) := e \cdot f$$

to be the intersection number of e and f. Since any two representatives of α are bordant, and intersection number is only defined up to bordism class, this map is well defined and we have that $\phi_{[e]}(\alpha) \in H^d(M)$.

[2]The conditions here are A cannot be of the form $A^2 = 0, A = mB$ and $g = 1$.

Definition 2.1.31

We say that a $(m-d)$ pseudocycle e is a **weak representative of** $\alpha_e \in H_d(M,\mathbb{Z})$ if

$$\mathbf{PD}(\Phi_{[e]}) = \beta$$

where **PD** is the isomorphism of Poincaré duality.

This formulation of representing homology classes gives us a way to translate between the evalution mapping and homology. In particular, it allows us to define something along the lines of "degree" without having a closed manifold. We have that the number of points in $\mathrm{ev}^{-1}(x)$ counted with sign is generically the multiplicity of

$$\mathbf{PD}(\Phi_{[e]}) \in H_1(M^l,\mathbb{Z}).$$

This gives us Theorem 2.1.5.

2.2 Taubes's Gromov invariant, with Caveats

We now have a basic understanding of the Moduli space of pseudoholomorphic curves that pass through a specified number of generic points. In order to define Taubes's Gromov invariant, we will count the number of curves that pass through these points in a specified way.

Definition 2.2.1

Given a symplectic manifold (M,ω) with compatible holomorphic structure J, a class A , and a set of $k(A = \frac{1}{2}(c_1(A) + A\cdot A)$ points Ω_k, define the set of J- holomorphic curves passing through Ω_k to be the set

$$\mathcal{H}(A,\Omega_k) := \{(\phi : \Sigma \to M \mid (*)\}/\sim$$

Where the conditions $(*)$ are

- The curves $\phi : \Sigma \to M$ are J holomorphic (possibly disconnected) curves.
- The points Ω_k are contained within the image of ϕ.
- The images of each connected component are distinct.
- $[\phi_*(\Sigma)] = A$

We say that two representations of the curve are similar (and write $\phi \sim \phi'$) if

- ϕ and ϕ' have the same images.
- The multiplicity of ϕ and ϕ' restricted to a connected component have the same multiplicity.

We denote the equivalence classes (ϕ,Σ).

Roughly, the space $\mathcal{H}(A,\Omega_k)$ denotes the unparameterized pseudoholomorphic curves that pass through the sets Ω_k. We should feel like we have a pretty good understanding of this set by thinking about the evaluation mapping from the previous section. There are only a few problems, most of them related to counting the multiplicity of certain mappings. We will need to cover this later, but let us look at the nice case first.

Definition 2.2.2

Given $(\phi,\Sigma) \in \mathcal{H}(A,\Omega_k)$, we say that it is **good** if $\phi|_{\Sigma_i}$ has multiplicity 1 (i.e. $\phi(\Sigma_i)\cdot(\Sigma_i) \le 0$).

There are some surprising facts about having good submanifolds representing a class of homology.

Theorem 2.2.3 Structure of $\mathcal{H}(A,\Omega_k)$

For generic ω-tame almost complex structure J on M, and $A \in H_2(M,\mathbb{Z})$, we have the following characterizations of $\mathcal{H}(A,\Omega_k)$.

1. Suppose that $(\phi,\Sigma) \in \mathcal{H}(A,\Omega_k)$ is good. Then :
   (a) Every $(\phi',\Sigma') \in \mathcal{H}(A,\Omega_k)$ is good.

(b) The image of the connected components are disjoint, that is $\phi(\Sigma_i) \cap \phi(\Sigma_j) = \varnothing$

(c) The multiplicity of $\phi|_{\Sigma_i}$ is one, unless $g(\Sigma_i) = 1$ and $\phi(\Sigma_i)$ has no self intersections.

(d) $\mathcal{H}(A)$ is zero dimensional.

2. If $(\phi, \Sigma) \in \mathcal{H}(A, \Omega_k)$, then the image $\phi(\Sigma_i)$ of every Σ_i such that $\phi(\Sigma_i) \cdot \phi(\Sigma_i) < 0$ is an embedded exceptional sphere, with multiplicity possibly greater than 1.

The proof of this theorem is mainly due to the evaluation map defined in the previous section. Refer to [11] for the proof of $ev^{-1}(x_1, \cdots, x_k)$ is compact for a generic choice of $(x_1, \cdots, x_k)$ and it is an invariant.

Now, we will present the definition of Taubes's Gromov invariant in a special case.

Definition 2.2.4 **Taubes's Gromov invariant**

Suppose $A \in H_2(M, \mathbb{Z})$ is a homology class such that $\mathcal{H}(A)$ only contains elements with components of multiplicity 1, then define

$$Gr(A) = \sum_{(\phi,\Sigma)\in\mathcal{H}(A)} \epsilon(\phi, \Sigma)$$

where $\epsilon(\phi, \Sigma)$ is the orientation of the point in $\mathcal{H}(A)$.

In the case where A can be parameterized by a good surface, then $Gr(A)$ is literally just counting with orientation the number of surfaces that represent this homology class passing through k generic points. From our work with the evaluation mapping before, we should expect this to be a finite number, as the space $\mathcal{H}(A, \Omega_k)$ should be identifiable with the preimage of the evaluation mapping. We therefore do not seem to be doing too poorly in the cases where $\mathcal{H}(A, \Omega_k)$ contains a good representative.

SW $\Rightarrow$ Gromov

The goal of this section is to show how one can use solutions the Seiberg-Witten equations to construct the pseudoholomorphic curves that are used in Definition 2.2 for the Gromov invariants. Taubes's paper [13] gives a method to construct pseudoholomorphic curves given a sequence of solutions to the Seiberg-Witten equations with varying pertubation parameter. The precise result of Taubes's paper is the following:

Theorem 3.0.5

Let M be a compact 4 manifold with symplectic form ω. Let $\tilde{P} \to M$ be a Spin^c structure with associated complex line bundle $\mathcal{L}$. Let Ω_i be a finite set of closed sets, and assume there is an unbounded sequence r_n for which $\mathcal{M}_{\mathrm{SW}_{r_n}}(\tilde{P})$ has a solution, where $\alpha^{-1}(0) \cap \Omega_i \neq \varnothing$. Then there exists a smooth compact pseudoholomorphic curve that passes through Ω_i with $\phi^*(\Sigma)$ equal to the Poincaré dual to $c_1(\mathcal{L})$.

In this exposition, we will show that the solutions of the Seiberg-Witten equations give us enough information to construct a 2 dimensional subspace of M. We will not show that the subspace is a submanifold, or even J-holomorphic.

3.1 Outline of Proof of Theorem 3

We would like to show that a sequence of the Seiberg-Witten solutions parameterized by r has a convergent subsequence. However, we are unable to do this for several reasons (the first being, that they do not live in the same space, unless we take parameterized configuration space.) In order to get these sections to converge, we will have to weaken our sense of convergence, and work with currents, instead of differential forms.

Definition 3.1.1
Currents

Let M be a differentiable manifold. A k-**current** on M is a linear functional from $\Omega^k_c(M)$. It comes equipped with the **mass norm**

$$|\mathcal{F}| := \sup_{\substack{\omega \in \Omega^k_c(M) \\ |\omega| \leq 1}} \mathcal{F}(\omega)$$

The reason to use currents is that they have much nicer convergence properties than differential forms. We should think of currents as being the differential form equivalent to distributions. Given a solution to the Seiberg-Witten equations we can translate the data about the connection into a much-better beahved current:

Definition 3.1.2
Associated Current

Let $(A, \alpha, \beta$ be a solution to the Seiberg-Witten equation. Define the **associated current** to be the current

$$\mathcal{F}_A(\mu) := \int \frac{i}{2\pi} F_A \wedge \mu.$$

Now let's show how these currents relate to the problem of constructing a surface from solutions to the Seiberg-Witten equations. Recall the Taubes Formulation for the Seiberg-Witten equations over a symplectic manifold

$$\begin{aligned} \sigma(u_0 \otimes \nabla_a \alpha) + D_A \beta &= 0 \\ F_a^+ &= \frac{-i}{8} r(1 - |\alpha|^2 + |\beta|^2) \cdot \omega + \frac{ir}{4}(\alpha\beta^* + \alpha^*\beta) \end{aligned}$$

where we use the decomposition $S_+ = (E \oplus (K^{-1} \otimes E))$ so that configurations (a, α, β) are given by

- a is a connection on E

- α is a section of E
- β is a section of $K^{-1} \otimes E$

In this equation $\alpha\beta^*$ and $\alpha^*\beta$ are sections of K and K^{-1} respectively. The perturbation parameter r comes from the section of perturbations

$$\eta = \frac{-ir}{4}\omega + F^+_{A_0} + i\eta_0$$

We therefore can define the **r-perturbed moduli space of solutions** to be $\mathcal{M}_{\mathrm{SW}_r}(\tilde{P})$. This is a smooth compact manifold for generic r. Let's assume the conditions of Theorem 3, which are a given

1. $\tilde{P} \to M$ be a Spin^c structure with associated complex line bundle $\mathcal{L}$
2. An unbounded sequence r_n of real numbers
3. Ω_i, a finite collection of closed sets.

with the conditions $\mathcal{M}_{\mathrm{SW}_{r_n}}(\tilde{P})$ has a solution, where $\alpha^{-1}(0) \cap \Omega_i \neq \varnothing$. Then we may construct associated currents to each of these specified solutions, $\mathcal{F}_n$.

Lemma 3.1.3

The sequence of currents is bounded with respect to the mass norm, that is

$$\|\mathcal{F}_n\|_M \leq \frac{1}{2\pi}\|F_{A_n}\|_{L^1} \leq [\omega] \cdot c_1(E) + c/r$$

where C is some constant.

This is proven by taking the a priori bounds for the curavture and massaging them to show that they have no r dependence. As the mass norm is bounded, we have the foloowing corollary

Corollary 3.1.4

There is a subsequence of the $\mathcal{F}_n$ which converges in the weak topology to a current $\mathcal{F}$.

This current should be thought of as the "limit" of solutions to the Seiberg-Witten solutions with increasing pertubation parameter. It shares many of the same properties as the $\mathcal{F}_n$

Claim 3.1.5 The current $\mathcal{F}$ satisfies the following properties

1. It is closed (i.e. $\mathcal{F}d(-) = 0$)
2. It is integral (i.e. $d\mu = 0$ and μ evaluates to an integer on all closed cycles implies $\mathcal{F}(\mu) \in \mathbb{Z}$).)
3. It is Poincaré dual to $c_1(\mathcal{L})$ (i.e. $\mathcal{F}(\mu) = [\mu] \cdot c_1(\mathcal{L})$.)

Our construction of $\mathcal{F}$ will give it an important geometric interpretation. The support of $\mathcal{F}$ will be a psuedoholomorphic curve. For an example of why how one might visualize this, consider the bump functions

$$\delta_r(x) = \frac{1}{re^{\left(\frac{x^2}{r^2}\right)}}$$

which become more concentrated with increasing parameter r. These converge to the Dirac delta distribution, n which "picks out" a point on the real line. In a similar fashion, by increasing the parameter r for the Seiberg-Witten solution, the currents $\mathcal{F}_n$ converge to $\mathcal{F}$ which picks out a 2 dimensional subspace.
The proof of 3 should be motivated from this Fundamental Proposition.

Proposition 3.1.6 (Support of $\mathcal{F}$)

The support of $\mathcal{F}$ is contained in a set C that has finite 2-dimensional Hausdorff measure.

We spend the rest of this section proving this proposition.

3.2 Constructing C

Here, we provide an outline for the construction of C, a subset of M with Hausdorff dimension 2 that contains the support of $\mathcal{F}$. The process for showing that this set has Hausdorff dimension 2 is as follows:

- Prove a bunch of useful inequalities about solutions to the Seiberg-Witten equations. We won't do this.
- (Subsection 3.2.1) Understand the energy function around zeros of solutions of the Seiberg-Witten equations.
- (Subsection 3.2.2) Another characteristic of solutions to the Seiberg-Witten equations is that all of the "interesting" parts of the spinor field die off at an exponential rate away from the zero set.
- (Subsection 3.2.3) Use the above two ideas to construct a certain number of balls that contain a large percentage of the energy associated to $\mathcal{F}_n$, and use that to make a statement about how much of the support of $\mathcal{F}_n$ is contained within those balls. Take the limit $\mathcal{F} = \lim_{n\to\infty} \mathcal{F}_n$, and use the earlier sections to show that the support of this current must be contained in a set of Hausdorff dimension 2 with finite measure.
- (Subsection 3.2.4) Finally, we motivate some of the structure of $\mathcal{F}$, and why we might expect the support to be a psuedoholomorphic curve by showing that $\mathcal{F}$ is of type $1-1$.

3.2.1 Energy of the Seiberg-Witten Equations

The Seiberg-Witten equations come with an associated energy functional, and understanding bounds on this energy gives us information on the solutions. Instead of trying to characterize the moduli space of SW equations with this energy functional, we will be interested in showing that the majority of the energy of a solution to the Seiberg-Witten equations is concentrated near the zero section of the equations; these bound will give topological data on the zeros of the Seiberg-Witten solutions.

Definition 3.2.1

Given a solution to the r-perturbed Seiberg-Witten equations $(A, (\alpha, \beta))$, and an open set U, define the **energy of α over U** by the integral

$$\mathcal{E}_U := \frac{r}{4}\int_U |1 - |\alpha|^2|$$

The center of Taubes's proof centers on the fact that the energy of the function cannot be too big; at the same time, one can show that there must be some large amount of energy concentrated around the zeros of α.

Proposition 3.2.2
Monotonicity

Let $(A, (\alpha, \beta))$ be a solution to the r-perturbed Seiberg-Witten equations. Then there is a constant $c \geq 1$ depending only on the choice of spin structure and Riemannian metric such that given:

- r is greater than c
- A geodesic ball of radius $\frac{1}{C} \geq s \geq \frac{1}{2r^{\frac{1}{2}}}$ centered at x

Then we have the following lower and upper bounds for the energy

- (Upper Bound) $\mathcal{E}_B \leq Cs^2$
- (Lower Bound) Provided that $|\alpha| < \frac{1}{2}$ we have that $\mathcal{E}_B \geq \frac{1}{C+1}s^2$.

This energy bound can be used to give produce a function whose derivatives are large when α is small, but whose overall size is bounded everywhere.

Proposition 3.2.3

Let $(A, (\alpha, \beta))$ be a solution to the r-perturbed Seiberg-Witten equations. Then there is a constant $c \geq 1$ depending only on the choice of spin structure and Riemannian metric such whenever $r \geq 1$ there exists a

smooth function $u : M \to \mathbb{C}$ such that

$$\begin{aligned} |u| &\leq c \\ \frac{1}{2} d^* du &\geq r \qquad \text{whenever } |\alpha| < \frac{1}{2} \\ |d^* du| &\leq cr \end{aligned}$$

The energy bound is not used directly to construct the function u, but it provides an argument for the structure of points where $\alpha \leq \frac{1}{2}$. In order to construct the function u, Taubes first constructs a covering for the set where $\alpha \leq \frac{1}{2}$ by balls using the bounds on the energy of α. The function is then created by smoothing out the characteristic equations for these balls. The properties of this covering are summarized in the following 3 lemmas:

Lemma 3.2.4 Given $(A, (\alpha, \beta))$ as before, there exists a constant c such that for $r \geq C$ and a fixed $r^{\frac{-1}{2}} < \rho < \rho_0/100$, we have the following:

- The set where $|\alpha| \leq \frac{1}{2}$ can be covered by $\frac{c}{\rho^2}$ balls of radius ρ.
- This set of covering balls $\{B_i\}$ can be chosen in such a way that if we replace them with balls of half radius, the set is disjoint.

This gives us a nice bound on the "size" of the support of the Seiberg Witten solutions. We also have that the balls capture a large portion of the energy— that is, by choosing a set with slightly larger α, we need not go out much farther from this set of balls to achieve this alpha value.

Lemma 3.2.5 Given $\lambda > 0$, there exists a constant c with dependence on λ , metric and choice of spin structure, with the following property. Let $(A, (\alpha, \beta))$ be a solution to the Seiberg Witten equations as before. Let V be the set of points where $|\alpha| \leq \frac{1}{2}$, and V' the set of points of size less that $\frac{3}{4}$. Cover the set V as in Lemma 3.2.1, and let B be a ball from this set. Let B' be a ball of radius $\frac{c}{r^{\frac{1}{2}}}$ centered at B. Then

$$\text{vol}((M - V') \cap B') \geq \lambda \, \text{vol}(B).$$

Finally, we have a bound on how clumped together the balls need to be.

Lemma 3.2.6 There is a constant c which depends only on the spin structure and metric such that if $B(n)$ is a ball of radius $\frac{n}{r^{\frac{1}{2}}}$, the number of balls from the covering in Lemma 3.2.1 whose centers lie in $B(n)$ is less than cn^2.

These bounds on covering balls will become useful later when we explore the creation of the set C.

3.2.2 Exponential Decay of Solutions

In this section, we will discuss about *Exponential decay estimates*. Before we proceed, we would like to emphasize the significance of *exponential decay estimates*. Taubes's paper relies heavily on the fact that the size of α, β and ∇'_A all drop off exponentially in value from the zero set of α and in the parametery r. This means that by increasing r, we can localize the non-trivial behavior of the Sieberg-Witten solutions to around the zero set of α.

Weitzenböck formula

We recall the isomorphism

$$S_+ = E \oplus (K^{-1} \otimes E).$$

Then every section ψ can be written of the form

$$r^{\frac{1}{2}} \cdot (\alpha u_0 + \beta)$$

where α is a section of E, $\alpha \cdot u_0$ is a section of E summand of S_+, and β is a section of the $K^{-1} \otimes E$ summand.

Let's recall the Seiberg-Witten equation

$$\begin{aligned} \not{\partial}_A \psi &= 0 \\ F_A^+ &= \frac{1}{4}\tau(\psi \otimes \psi^*) \end{aligned}$$

For *Exponential decay estimates*, we will mainly deal with inequalities induced from the first equation of the Seiberg-Witten equation. Since the Dirac operator is defined by

$$\not{\partial}_A \psi := \sigma(\nabla_A \psi) : S_+ \to S_+ \otimes T^*X \to S_-,$$

we can apply $\not{\partial}_A$ twice to ψ. Then, we get $\not{\partial}_A^2 \psi = 0$ for every solution of Seiberg-Witten equation. After some computations (also using $F_A^+ = \frac{1}{4}\tau(\psi \otimes \psi^*)$), the equation ends up with an equation, *Weitzenböck formula*,

$$\nabla_A^* \nabla_A \psi + \frac{r}{4}\psi + \frac{1}{2} \cdot c_+(F_A^+)\psi = 0. \tag{3.1}$$

(Here, ∇_A^* is the adjoint of ∇_A.) The equation above is, in fact, the whole thing we need in order to prove results about *Exponential decay estimates*.

Differential Inequalities

We will skip precise calculation process but give results and simple explanations about them. In this section we will use c and c_n to stand for a variety of different constants. The first inequality we would like to get is

$$\begin{aligned} &\frac{1}{2}d^*d|\nabla_a\alpha|^2 + |\nabla_a\nabla_a\alpha|^2 + \frac{r}{4}|\alpha|^2|\nabla_a\alpha|^2 \\ &\quad \le c \cdot (r \cdot (1 - |\alpha|^2) + 1)|\nabla_a\alpha|^2 \\ &\quad + c \cdot |\nabla_a\alpha| \Big(\ (1 + r|\beta|) \cdot |\nabla'_A\beta| + (r|\alpha|^2 + 1)|\beta| + |\nabla'_A\nabla'_A\beta| \ \Big). \end{aligned}$$

First, project the equation $\not{\partial}_A^2 \psi = 0$ onto the E summand of $S_+ = E \oplus (K^{-1} \otimes E)$. Then, take the action by ∇_a and inner product with $\nabla_a\alpha$. One thing to be noted is that c only depends on the Riemannian metric.

Using an easy inequality that $2|f||g| \le r|f|^2 + \frac{1}{r}|g|^2$, we can convert this inequality to

$$\begin{aligned} &\frac{1}{2}d^*d|\nabla_a\alpha|^2 + |\nabla_a\nabla_a\alpha|^2 + \frac{r}{4}|\alpha|^2|\nabla_a\alpha|^2 \\ &\qquad \le \left(c_1 \cdot r(1 - |a|)^2 + \frac{r}{16}\right)|\nabla_a\alpha|^2 \\ &\qquad + c_1 \cdot r|\beta|^2 + c_1 \cdot |\nabla'_A\beta|^2 + \frac{32}{r}|\nabla'_A\nabla'_A\beta|^2 \end{aligned} \tag{3.2}$$

In a very similar sense, project the Eq 3.1 onto the $K^{-1} \otimes E$ summand and then take the action by ∇'_A and the inner product with $\nabla'_A\beta$. Then we get

$$\begin{aligned} &\frac{1}{2}d^*d|\nabla'_A\beta|^2 + |\nabla'_A\nabla'_A\beta|^2 + \frac{r}{4}|\alpha|^2|\nabla'_A\beta|^2 \\ &\quad \le c_2 \cdot (r(1 - |\alpha|^2) + 1) \cdot |\nabla'_A\beta|^2 \\ &\quad + c_2|\nabla'_A\beta| \Big(\ (1 + r|\alpha||\beta|) \cdot |\nabla_a\alpha| + r|\alpha||\beta|^2 + |\nabla_a\nabla_a\alpha| \ \Big). \end{aligned}$$

For the next step, we again use the simple inequality we have used before. Then, we get

$$\begin{aligned} &\frac{1}{2}d^*d|\nabla'_A\beta|^2 + |\nabla'_A\nabla'_A\beta|^2 + \frac{r}{4}|\alpha|^2|\nabla'_A\beta|^2 \\ &\qquad \le (c_3 r(1 - |\alpha|^2) + \frac{r}{16})|\nabla'_A\beta|^2 + c_3|\beta|^2 \\ &\qquad + c_3|\nabla_a\alpha|^2 + \frac{32}{r}|\nabla_a\nabla_a\alpha|^2 \end{aligned} \tag{3.3}$$

Now, we have got two inequalities which we want to add to each other after some scalar multiplication. Calculating $(3.2) + \frac{r}{32} \times (3.3)$, we get an inequality

$$\frac{1}{2}d^*dy + \frac{r}{4}|\alpha|^2 y \leq \left(c_4 r(1-|\alpha|)^2 + \frac{r}{16}\right) y + c_4 r|\beta|^2 \tag{3.4}$$

where $y := |\nabla_a \alpha|^2 + \frac{r}{32}|\nabla'_A \beta|^2$.

Before we get into the main step, we shall induce an inequality from (3.1) again. Simply, take the inner product of (1) with β.

$$\frac{1}{2}d^*d|\beta|^2 + |\nabla_A \beta|^2 + \frac{r}{4}|\beta|^2(1+|\alpha|^2+|\beta|^2) - c_1|\beta|^2 \leq c_2|\beta||\nabla_a \alpha|.$$

With $r > 16c_1$, this inequality (also using the inequality $2|f||g| \leq r|f|^2 + \frac{1}{r}|g|^2$) can be simplified as

$$\frac{1}{2}d^*d|\beta|^2 + \frac{r}{4}|\alpha|^2|\beta|^2 \leq -\frac{r}{8}|\beta|^2 - |\nabla_A \beta|^2 + \frac{c_3}{r}|\nabla_a \alpha|^2. \tag{3.5}$$

Finally, we get an inequality (as a result of $(3.4) + \frac{r^2}{16c_3} \times (3.5)$)

$$\frac{1}{2}d^*dy_1 + \frac{r}{4}|\alpha|^2 y_1 \leq \left(c_4 r(1-|\alpha|^2) + \frac{r}{8}\right) y_1$$

where $y_1 := |\nabla_a \alpha|^2 + \frac{r}{32}|\nabla'_A \beta|^2 + \frac{r^2}{16c_3}|\beta|^2$.

Here, the reason we have *Exponential decay* outside of some $\alpha^{-1}(0)$ is that we can make $|\alpha| < 1$ very close to 1 for all points $x \in X$ such that $\mathrm{dist}(x, \alpha^{-1}(0)) \geq c_5 r^{-\frac{1}{2}}$ for some constant c_5. (This is a general fact for a solution of the Seiberg-Witten equation.) Then this gives

$$\frac{1}{2}d^*dy_1 + \frac{r}{32}y_1 \leq 0$$

for all $x \in X$ such that $\mathrm{dist}(x, \alpha^{-1}(0)) \geq c_5 r^{-\frac{1}{2}}$.

Now, define a function $H_y(x)$ for $x, y \in X$ as

$$\frac{1}{\mathrm{dist}(x,y)^2}\exp\left(-\frac{1}{c}r^{\frac{1}{2}}\mathrm{dist}(x,y)\right) + c \cdot \exp\left(-\frac{1}{c}r^{\frac{1}{2}}\right).$$

The main role of this function comes from the fact that

$$\frac{1}{2}d^*dH_y + \frac{r}{32}H_y \geq 0$$

as long as $\mathrm{dist}(x,y) \geq c \cdot r^{-\frac{1}{2}}$. We now define

$$h := \sum_i H_{x_i}$$

where $\{x_i\} \subset \alpha^{-1}(0)$ is maximal such that $B_{r^{-\frac{1}{2}}}(x_i)$'s are disjoint. Then, the behavior of $h(x)$ is bounded by

$$h(x) \leq c_6 \cdot r \exp(-\frac{1}{c}r^{\frac{1}{2}} \cdot \mathrm{dist}(x, \alpha^{-1}(0)))$$

where $\mathrm{dist}(x, \alpha^{-1}(0))$. Now, we can find a constant c_8 such that

$$\frac{1}{2}d^*d(y_1 - c_8 \cdot h) + \frac{r}{32}(y_1 - c_8 \cdot h) \leq 0$$

where $\mathrm{dist}(x, \alpha^{-1}(0)) \geq c_5 \cdot r^{-\frac{1}{2}}$ and on the boundary where $\mathrm{dist}(x, \alpha^{-1}(0)) = c_5 \cdot r^{-\frac{1}{2}}$, $y_1 - c_8 \cdot h \leq 0$. So, this gives a differential inequality with some initial boundary condition. Hence, y_1 is bounded by a scalar multiple of h and h has *Exponential decay*. Therefore, y_1's each term has Exponential decay estimates.

Proposition 3.2.7

For a fixed Spin^c structure $\tilde{P}$, there exists a constant c such that for any solution $(A,(\alpha,\beta)$ to the Seiberg-Witten equations, we have the following bounds:

$$r(1-|\alpha|^2),\ \ r^{2/3}|\beta|,\ \ |F_\alpha|,\ \ r^{1/2}|\nabla_A\alpha|,\ \ r|\nabla'_A\beta| \leq cr\exp\left(-\frac{r^{1/2}}{c}\mathrm{dist}(x,\alpha^{-1}(0))\right)$$

3.2.3 Covering the Solution with Balls

With the knowledge that the zero section of the solutions can be covered by a set of balls of radius ρ that grow in number as ρ^2, (Lemma 3.2.1), and that the majority of the interesting behavior of the Seiberg Witten equations is contained in a small volume around the zero section (Subsection 3.2.2), we are ready to prove that the support of the current $\mathcal{F}$ is contained in a set of Hausdorff Dimension 2.
For each N define the set $\Lambda_n(N)$ to be the set of $c(256)^N$ balls with radius $r\cdot(16)^{-N}$ centered at the points that cover $\alpha_n^{-1}(0)$. We may assume that the center of these balls are contained in $\alpha_n^{-1}(0)$. We may then take $U(N)$ to be the set of $c(256)^N$ balls of radius $4\cdot(16)^{-N}$ that eventually contain all of the $\Lambda_n(N)$ for sufficiently large n. These $U(N)$ are nested inside of each other. Define the set

$$C := \bigcap_N U(N)$$

Notice that C here is independently defined of the choices that we make for these balls.
It is clear from construction that C has finite 2-dimensional Hausdorff measure. But C also has the following nice properties:

Lemma 3.2.8

The set C has the following properties

- It has finite 2-dimensional Hausdorff measure.
- There is a constant C such that for all $x \in C$, and B a ball of radius ρ, then the 2-dimensional Hausdorff measure of $C \cap B$ is at least $C\rho^2$.
- C contains the support of $\mathcal{F}$

While the full details of this proof are rather technical, the basic idea is as follows:
The Hausdorff dimension of C follows simply from the construction of C. Recall that the Hausdorff dimension is simply the smallest d such that the d-dimensional Hausdorff measure of C is still finite, and the d-Hausdorff measure is given by

$$H^d(C) = \lim_{r_0\to 0}\inf\left(\sum_{i=1}^{\infty} r_i^d\right)$$

where the infinum is over all covers of C by balls of radius $r_i < r_0$. Since the number of balls in $U(N)$ grows as 256^N, while the radius of the balls shrink as 16^{-N} we have that the $d = 2$ will still produce a finite 2-Hausdorff measure of the set as $256 = 16^2$.
The monotonicity formula Proposition 3.2.1 shows that the majority of the energy of a solution must be near the zero section. This shows that balls centered at C have intersection with C of "proper" 2 dimensional area.
Two show that C contains the support of $\mathcal{F}$, consider a bump μ supported outside of C C. Then for some N, μ will be disjoint from $U_n(N)$ for all large n. Since the balls in $U_n(N)$ were designed to decrease at subexponential rate, while the size of the curvature grows at an exponential rate, we have,

$$|\mathcal{F}_n(\mu)| \leq c\cdot r\cdot exp\left(r_n^{\frac{1}{2}}16^{-N}\right)\cdot\sup_x|\mu(x)|.$$

This shows that the support of $\mathcal{F}$ does not contain the support of μ, proving the lemma.

3.2.4 Further Properties of C

While we will not show that C is a psuedoholomorphic curve, we will provide some indications of why we should expect it to be one. We can show that C is in some way "J-invariant." On the space of 2 forms there

is an involution defined by multiplication by J. By taking duals, we get an involution of J on the space of currents. We say that a current if **type** $1-1$ if it is invariant under this J involution.

Lemma 3.2.9 The current $\mathcal{F}$ is of type $1-1$.

Proof. Let us test a form μ against $\mathcal{F}_n$. We may assume that μ is locally supported around C, as we have shown that the behavior of μ away from C does not $\mathcal{F}_n(\mu)$ as $n \to \infty$. We have that

$$\mathcal{F}_n(\mu - J\mu)$$

is dependent on only the terms not fixed by J. These are the terms in $F^+_{A_n}$ which are linear in β_n and β_n^*. It can be shown that

$$|\beta_n|^2 \leq \frac{c}{r}\left(1 - |\alpha|^2 + \frac{c}{r_n^2}\right)$$

where c is some constant. Therefore, the terms of $F^+_{A^n}$ that are linear are bounded by

$$|\mathcal{F}_n(\mu - J\mu)| \sim \||\beta_n||\mu|\|_{L^1}$$

Using Hölder's inequality and the fact that μ has support close to C

$$\begin{aligned}
&\leq \|\chi_{U(N)}\beta_n\|_{L^1}\|\mu\|_{L^\infty} \\
&\leq \sup|\mu| \cdot \|\chi_{U(N)}\|_2 \cdot \|\beta_n\|_{L^2} \\
&\leq c \sup|\mu| \cdot (\mathrm{vol}(U(N))^{\frac{1}{2}} \cdot \|r_n(1 - |\alpha_n|^2)\|_{L^1}
\end{aligned}$$

Becuase of compactness we have that the first term is bounded. The second term goes to zero. The last term in this is bounded by the energy bound above. Taking the limit as $n \to 0$, we see that $\mathcal{F}(\mu - J\mu) = 0$. This proves that $\mathcal{F}$ is of type $1-1$. ∎

This places fairly strong conditions on the support of $\mathcal{F}$, suggesting that it has some kind of J-invariant properties. While we will not go so far as to actually show that the tangent space to the support is J invariant, we should be very hopeful that this is the case from this result.

3.2.5 Other Properties of C

This exposition only covers a small portion of Taubes's proof. From here, Taubes goes on to prove that the set C is not just a 2 dimensional set, but a psuedoholomorphic curve. The techniques used in this part are rather technical, but here is an outline of the methodology for the rest of the paper:

- One of the constructions that we did not cover is describing the "local structure of $\alpha^{-1}(0)$." By using a sufficently small trivialization, Taubes compares the zero sets of the Seiberg-Witten solutions over M to the zeros of the the Seiberg-Witten solutions over $\mathbb{C} \times \mathbb{C}$. While these generally do not have the same structure, they are similar enough that we can pay a very small price for working in one or the other.
- The structure of C is studied by looking at its intersections with Psuedoholomorphic disks. One uses an argument similar to the $(1-1)$-ness of $\mathcal{F}$ to show that the set C intersects certain disks positively. The claim is that for every disk $\sigma : D \to M$ the assignement

  $$I(\sigma) := \lim_{n\to\infty} \int_D \frac{i}{2\pi}\sigma^* F_{A_n}$$

  has the following propeties :

  1. If $\sigma(D)$ doesn't intersect C, $I(\sigma) = 0$
  2. $I(\sigma)$ is defined up to homotopy.

3. I respects muliplicity of covering, that is if $\theta_k : D \to D$ is the degree k mapping, then $I(\sigma\theta_k) = kI(\sigma)$.
4. $I(\sigma)$ is additive across disjoint unions.
5. If σ is pseudoholomorphic, then $I(\sigma) > 0$.

- Taubes then shows that 2 dimensional surfaces that correspond to positive cohomology assignements (given by the conditions above) necessarily correlate to pseudoholomorphic curves. Furthermore, the positive cohomology assignment gives us the multiplicity that we should count the curve with in the Gromov-Taubes invariant.

3.3 Where to Proceed from here

This theorem had some interesting implications about Gromov type invariants. We summarize some of the major results discussed in [17] here.

Theorem 3.3.1 Let M be an oriented 4-manifold with $b_2^+ > 1$. Then the Gromov Invariants are not dependent on choice of symplectic form (other than a choice of canonical line bundle.)

This follows trivially from the fact that the Seiberg-Witten equations are diffeomorphism invariant. This is rather remarkable, as the Gromov invariants look like they have a great deal of dependence on choice of symplectic structure.

Theorem 3.3.2 Let X be a compact oriented 4-manifold with $b_2^+ > 1$ and symplectic form ω. Let $e \in H^2(\mathbb{Z})$ with $SW(e) \neq 0$. Then the Poincaré dual to e is represented by the fundamental class of an embedded, symplectic curve with ginus $1 + e \cdot e$.

This means that elements of homology with non-trivial Seiberg-Witten invariant are representatble by nice symplectic curves.

Theorem 3.3.3 The manifold $\mathbb{CP}^2$ has unique (up to symplectomorphism) symplectic structure.

The proof of this relies on the fact that the Seiberg-Witten invariants of $\mathbb{CP}^2$ vanish due to positive curvature of $\mathbb{CP}^2$. This allows the construction of a certain type of 2-sphere. A theorem of gromov states that such a two sphere impleis the existence of symplectomorphism to the standard Kähler form.
The technique for proving that $\mathbb{CP}^2$ has unique symplectic structure can be used to show that other manifolds admit no symplectic structure at all. Finally, this proof gives us an idea of where to start to define "Gromov-type" invariants for non-symplectic manifolds. There exists manifolds with non-zero Seiberg-Witten invariants which do not admit symplectic structure. If M does not admit a symplectic structure, we can pick ω to be a nontrivial ASD harnomic 2 form, which vanishes at points of M. The theorem suggest that there is still a theory of psuedo-holomorphic curves for degenerate ω given by the Seiberg-Witten equations.

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